Joint Confidence Regions for Rankings Based on Correlated Estimates

Shaine Rosewel Paralis Matala

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1 Introduction

Ranked data are commonly of interest because they allow readers to compare populations based on ranked quantities. For example, top universities across the globe may be identified based on their institutional performance indicator; states may receive appropriate intervention according to their relative rank based on average travel times to work; and senatorial candidates who are likely to win seats can be predicted by public opinion polls prior to elections. Whenever data on the entire population are unavailable, the ranks are based on estimates rather than the unknown parameters. Consequently, their overall uncertainty—expressed through joint confidence intervals—should also be quantified. Marginally, these intervals provide information on the possible range of each rank while jointly, they facilitate comparing all ranks simultaneously rather than reporting them in isolation.

Several studies have addressed this concern through different techniques. Some approaches like those by Klein et al. (2020), Mohamad et al. (2019), Mogstad et al. (2024), Andersson et al. (1998) and Lyhagen & Ahlgren (2020), relied solely on the estimates and their standard errors, constructing joint confidence intervals either for the estimated quantities or directly for the ranks themselves. Others fit models in which institutional differences are modeled explicitly via latent effects, thereby accounting for the dependence among student observations induced by the multilevel structure (Spiegelhalter, 1996). Miller (2009) developed a bootstrap algorithm that allows for the assumption of independence in the estimators despite its potential violation while Bazylik et al. (2025) proposed a bootstrap procedure that explicitly accounts for the dependence in estimators that are multinomially distributed.

Assuming independence when constructing joint confidence regions for estimators that are, in fact, correlated may lead to overly conservative and thus

wider intervals, implying greater uncertainty—contrary to what is desired. The more general setting, which to our knowledge has not yet been explored, allows for a certain degree of correlation among the estimates to be ranked. This thesis develops a procedure capable of handling such dependencies while maintaining coverage close to the nominal level and producing relatively narrow joint confidence intervals. The proposed methodology uses only the observed estimators and their corresponding standard errors. Although it also employs a parametric bootstrap—commonly used to estimate overall rank uncertainties (e.g., Andersson et al. (1998), Mogstad et al. (2024), Bazylik et al. (2025))—our implementation differs from these existing approaches — Mogstad et al. (2024) and Bazylik et al. (2025) obtained bounds for the simultaneous confidence sets which depend on quantiles from the distribution of the maximum (over k) of the differences (See Section 2.3.2, instead of just the estimators while Andersson et al. (1998) implemented it to obtain a calculated measure as discussed in Section 2.3.1.

1.1 Objective

This research builds upon Klein et al. (2020)'s methodology by extending the set of joint confidence intervals used to capture uncertainty in overall rankings. In particular, it intends to:

- Develop a procedure to construct joint confidence intervals for the ranks and the ranked parameters when the estimates to be ranked may be correlated.
- Evaluate the performance of the proposed approaches under various parameter settings.
- Compare the results of Algorithm 1 to Klein's joint confidence regions that uses Bonferroni and independence assumption, in terms of coverage and tightness measures.
- Apply the proposed approaches to a real-life example.

1.2 Significance

In order to obtain joint confidence sets for overall ranks, Klein et al. (2020) estimates confidence intervals for the unknown parameters, with a joint coverage probability of at least $1 - \alpha$. Their goal is to produce confidence intervals that collectively produce a small difference between the upper and the lower bound to yield tighter joint uncertainty for ranks. In the same study, they used individual confidence intervals of the form $\hat{\theta}_k \pm z_{\alpha/2} \times \sigma_k$ for the population parameters $\theta_1, \theta_2, \ldots, \theta_K$, assuming that the estimates of these parameters $\hat{\theta}_1, \hat{\theta}_2, \ldots, \hat{\theta}_K$ are independently distributed. This approach, while simple, is no longer applicable to the case where $\hat{\theta}_k$s may be correlated. In many cases, assuming independence in the presence of correlated quantities leads to conservative confidence intervals resulting in wider intervals which imply a higher uncertainty in overall ranks.

In the Philippines, senators are elected using Multiple Non-transferable Vote system (MNTV)—where candidates are voted for individually regardless of partisan membership and alliances (Ravanilla & Hicken, 2023). This applies to pre-election polls where each voter X_i selects at most 12 from a pool of K candidates. We let the data $\mathcal{X}$ be a set of n vectors of length K, $\mathcal{X} = \{X_1, X_2, \ldots, X_n\}$, where $X_i = (X_{i1}, X_{i2}, \ldots, X_{iK})$ and $X_{ik} \sim$ Bernoulli(p_k) where p_k is the probability that voter i selects candidate k for $k \in \{1, 2, \ldots, K\}$ and $i \in \{1, 2, \ldots, n\}$. $X_{ik} = 1$ if voter i selects candidate k and 0 otherwise. Public opinion polling bodies report their results by ranking the proportion of votes $\hat{\theta}_k$ garnered by the K candidates from the highest to the lowest, stating individual uncertainty to emphasize who can potentially make it to the top 12. Figure 1.2.1 is an example of results from Pulse Asia Research, Inc. (2025); this shows that 19 out of the 64 senatorial candidates have statistical chances of winning a seat as of a week prior to the previous 2025 elections. The shaded rows correspond to the ultimate winners. Pulse Asia Inc. notes their surveys are conducted with a 95% confidence level, with confidence intervals calculated using 2.8% margin of error for a sample size of 1,200. To obtain the estimates, each response is weighted to

compensate for the unequal selection probabilities in the sample design.

Candidate	Voting Preference (6-9 May 2025)	Rank
Go, Bong Go (PDPLBN)	59.5	1
Tulfo, Erwin (LAKAS)	48.7	2-4
Sotto, Tito (NPC)	44.1	2-4
Dela Rosa, Bato (PDPLBN)	42.9	2-5
Lapid, Lito (NPC)	38.0	4-8
Cayetano, Pia (NP)	34.3	5-12
Lacson, Ping (IND)	33.1	5-13
Aquino, Bam (KNP)	32.8	5-15
Bong Revilla, Ramon Jr. (LAKAS)	32.3	6-15
Binay, Abby (NPC)	30.5	6-16
Marcoleta, Rodante (IND)	28.9	6-18
Tulfo, Ben (IND)	28.9	6-19
Pacquiao, Manny Pacman (PFP)	28.4	7-19
Marcos, Imee R. (NP)	27.6	8-19
Villar, Camille (NP)	27.6	8-19
Salvador, Phillip Ipe (PDPLBN)	26.5	10-19
Pangilinan, Kiko (LP)	25.1	11-20
Revillame, Willie Wil (IND)	24.1	11-20
Abalos, Benhur (PFP)	23.7	12-20

Figure 1.2.1: Candidates in Contention in May 2025 Pulso ng Bayan Survey by Pulse Asia Research Inc.

In weak-party systems, candidates who belong to the same political alliance or ticket commonly co-occur in ballots and hence perform with similarity (David & Legara, 2015). As shown in Figure 1.2.1, the competitive list is dominated by ten out of 11 (all except Tolentino) candidates under *Alyansa para sa Bagong Pilipinas*, which was actively campaigned by the incumbent president at the time. Four candidates (Go, Dela Rosa, Salvador, and Marcoleta) are associated with the Duterte-backed alliance *DuterTen*, while two (Pangilinan and Aquino) belong to the opposition-backed *KiBam*. The remaining candidates—Marcos, Revillame, and B. Tulfo—do not formally belong to any alliance but benefit from strong name recall. This alliance-based pattern is consistent with dependence in individual voting behavior.

Because each voter is allowed to vote for up to 12 candidates, votes cast by the same individual are not independent. This dependence is reflected at the aggregate level through the proportions of total votes received by candidates, resulting in correlated proportion estimates. Accounting for these dependencies can lead to rank estimates with reduced overall uncertainty

and, consequently, shorter simultaneous confidence intervals. In particular, the range of plausible ranks for each candidate may be narrowed, depending on the degree of correlation among candidates' vote shares.

1.3 Scope and Limitations

This study proposes methodologies to construct joint confidence regions to quantify uncertainty in overall ranks, building upon the main results of Klein et al. (2020). Unlike the study of Klein et al. (2020), however, this thesis addresses the case where the parameter estimates may be correlated. Moreover, this study also proposes a methodologies to construct joint confidence regions for the ordered parameters. It applies the parametric bootstrap in implementing the procedures, with a strong emphasis on maintaining the desired coverage. However, several limitations must be acknowledged. First, the framework assumes that the data are generated from a multivariate normal distribution. Second, a limited set of correlation structures is examined to illustrate how different dependence assumptions may affect the resulting joint confidence sets. That it, we do not discuss how to estimate the correlation matrix from the data.

2 Related Literature

2.1 Ranking Problem

In the problem of estimating ranks of several unknown real-valued parameters $\theta_1, \theta_2, \ldots, \theta_K$, it is desired to rank these K parameters from smallest to largest, $\theta_{(1)} < \theta_{(2)} < \ldots < \theta_{(K)}$. Let $r_1, r_2, \ldots, r_K$ be the true unknown ranks of $\theta_1, \theta_2, \ldots, \theta_K$. For example, if $\theta_{(2)} = \theta_4$, then $r_4 = 2$. Our first goal is to derive a joint confidence region for $\theta_1, \theta_2, \ldots, \theta_K$ and thereby obtain joint confidence intervals for the $r_1, r_2, \ldots, r_K$ using a result from Klein et al. (2020). In deriving a joint confidence region for $\theta_1, \theta_2, \ldots, \theta_K$, we will make use of the corresponding point estimates $\hat{\theta}_1, \hat{\theta}_2, \ldots, \hat{\theta}_K$. Overall uncertainty highlights that in pre-election polls, reported candidate rankings that seem decisive may in fact be much less stable once uncertainty is considered, and hence it is possible that these rankings do not align with the actual election outcomes.

Our second goal is to derive a joint confidence region for the ordered parameters $\theta_{(1)}, \theta_{(2)}, \ldots, \theta_{(K)}$. In the context of senatorial elections where θ_k denotes the proportion of votes earned by a candidate, confidence intervals on ordered parameters help determine if a candidate's lead is statistically significant. If the confidence intervals for two candidates' support levels do not overlap, it is highly likely (at the specified confidence level) that the candidate with the higher point estimate is the true winner.

A mathematical definition of r_k is as follows:

$$r_k = \sum_{j=1}^{K} I(\theta_j \leq \theta_k) = 1 + \sum_{j:j \neq k} I(\theta_j \leq \theta_k), \qquad \text{for } k = 1, 2, \ldots, K. \quad (2.1.1)$$

Let $\hat{r}_k$ denote the estimated rank of the kth observation. This can be

computed determined based on

$$\hat{r}_k = 1 + \sum_{j:j\neq k} I(\hat{\theta}_j \leq \hat{\theta}_k), \qquad \text{for } k = 1, 2, \ldots, K. \tag{2.1.2}$$

In this chapter, we review the various approaches to solve this problem of finding a joint confidence region for $r_1, r_2, \ldots, r_K$.

2.2 Klein's Joint Confidence Region for Overall Ranking Uncertainty

The study of Klein et al. (2020) assumes that $\hat{\theta}_k \sim N(\theta_k, \sigma_k^2), k = 1, 2, \ldots, K$ where θ_k is unknown but σ_k^2 is known. We now describe their methodology.

Suppose that for each $k \in \{1, 2, \ldots, K\}$ there exists values L_k and U_k such that

$$\theta_k \in (L_k, U_k), \qquad U_k, L_k \in \mathbb{R} \tag{2.2.1}$$

Moreover, define the quantities I_k, Λ_{Lk}, Λ_{Rk}, Λ_{Ok} as follows:

$$\left.\begin{array}{c} I_k = \{1, 2, \ldots, K\} - \{k\}, \\ \Lambda_{Lk} = \{j \in I_k : U_j \leq L_k\}, \\ \Lambda_{Rk} = \{j \in I_k : U_k \leq L_j\}, \\ \Lambda_{Ok} = \{j \in I_k : U_j > L_k \text{ and } U_k > L_j\} = I_k - \{\Lambda_{Lk} \cup \Lambda_{Rk}\} \end{array}\right\}$$

which implies the following:

1. $j \in \Lambda_{Lk} \leftrightarrow (L_j, U_j) \cap (L_k, U_k) = \emptyset$ and (L_j, U_j) lies to the left of (L_k, U_k);
2. $j \in \Lambda_{Rk} \leftrightarrow (L_j, U_j) \cap (L_k, U_k) = \emptyset$ and (L_j, U_j) lies to the right of (L_k, U_k);
3. $j \in \Lambda_{Ok} \leftrightarrow (L_j, U_j) \cap (L_k, U_k) \neq \emptyset$
4. $\Lambda_{Lk}, \Lambda_{Rk}$, and Λ_{Ok} are mutually exclusive, and $\Lambda_{Lk} \cup \Lambda_{Rk} \cup \Lambda_{Ok} = I_k$

Let $|A|$ denote the number of elements in a set A. If the condition in (2.2.1) holds, the main result from Klein et al. (2020) gives a range for the

value of r_k for each $k \in \{1, 2, \ldots, K\}$ as follows:

$$r_k \in \{|\Lambda_{Lk}| + 1, |\Lambda_{Lk}| + 2, |\Lambda_{Lk}| + 3, \ldots, |\Lambda_{Lk}| + |\Lambda_{Ok}| + 1\} \tag{2.2.2}$$

Note that the number of elements in the range given in (2.2.2) is $|\Lambda_{Ok}| + 1$. Since the smaller difference between U_k and L_k leads to a smaller $|\Lambda_{Ok}|$, narrower confidence intervals for $\theta_1, \theta_2, \ldots, \theta_K$ are desirable.

Suppose that for random quantities L_k and U_k the event defined in (2.2.1) satisfies the following probability condition:

$$P\left[\bigcap_{k=1}^{K} \{\theta_k \in (L_k, U_k)\}\right] \geq 1 - \alpha, \tag{2.2.3}$$

then, by the result of Klein et al. (2020), it also follows that

$$P\left[\bigcap_{k=1}^{K} \{r_k \in \{|\Lambda_{Lk}| + 1, |\Lambda_{Lk}| + 2, |\Lambda_{Lk}| + 3, \ldots, |\Lambda_{Lk}| + |\Lambda_{Ok}| + 1\}\}\right] \geq 1-\alpha. \tag{2.2.4}$$

Due to the assumption of normality on $\hat{\theta}_k$ as well as the fact that σ_k is assumed known, Klein et al. (2020) set the confidence intervals (L_k, U_k) for θ_k to be of the form $\hat{\theta}_k \pm t \times \sigma_k$ for $k \in \{1, 2, \ldots, K\}$. One may use the Bonferroni approach to choose t and call the resulting confidence region $\mathfrak{R}_{\text{bonf}}$. If such an approach is used, the choice of t that would satisfy (2.2.3) is $t = z_{\alpha/2K}$. Another choice of t is one that exploits the independence assumption on $\hat{\theta}_k$; denote the associated confidence region with $\mathfrak{R}_{\text{ind}}$. Such a choice is given by $z_{\gamma/2}$ where $\gamma = 1 - (1 - \alpha)^{1/K}$. To see why this is the case, note that if we define Y_k as follows:

$$Y_k = \frac{\left(\hat{\theta}_k - \theta_k\right)}{\sigma_k} \sim N(0, 1), \quad k = 1, 2, \ldots, K \tag{2.2.5}$$

then we get

$$
\begin{aligned}
\prod_{k=1}^{K} P\left(|Y_k| \leq t\right) &= \prod_{k=1}^{K} P\left(\hat{\theta}_k - t \cdot \sigma_k \leq \theta_k \leq \hat{\theta}_k + t \cdot \sigma_k\right) \\
&= \prod_{k=1}^{K} 1 - [1 - (1-\alpha)^{1/K}] \qquad (2.2.6) \\
&= 1 - \alpha.
\end{aligned}
$$

2.3 Alternative Approaches for Ranking Uncertainty

While Klein's approach provides one framework for constructing joint confidence regions for ranks, several other studies have explored related problems using different formulations or assumptions. These alternative methods vary in whether they account for dependence structures, rely on model-based estimation, or use resampling techniques such as the bootstrap.

2.3.1 Calculated Measure

Other studies with similar concern include that of Andersson et al. (1998) who suggest the use of a statistic C which quantifies the number of positions ranked parameters would on average change their order due to random variation. They calculate the measure using a bootstrap approach. Working on risk ratios p_k of K units, Andersson et al. (1998) draw B bootstrap proportions $\hat{p}_k^*$ from (2.3.1):

$$
\hat{p}_{bk}^* \sim N\left(\hat{p}_k, \frac{\hat{p}_k(1-\hat{p}_k)}{n_k}\right), \quad k = 1, 2, \ldots, K;\ b = 1, 2, \ldots, B \qquad (2.3.1)
$$

and $\hat{p}_k$ and n_k are the sample proportion and sample size, respectively, corresponding to the kth unit. For each bootstrap iteration b, they sort the set $\hat{p}_{b1}^*, \ldots, \hat{p}_{bK}^*$ to get the corresponding ranks $r_{bk}^*, k = 1, 2, \ldots, K$. The difference d_{bk} between the original and bootstrap rank is then calculated as $d_{bk} = |\hat{r}_k - \hat{r}_{bk}^*|$, where $\hat{r}_k$ is the rank of $\hat{p}_k$ in the set $\{\hat{p}_1, \hat{p}_2, \ldots, \hat{p}_K\}$. In turn, $\bar{d}_k$ is obtained by taking the average of d_{bk} across the bootstrap

samples. Finally, the overall measure C is calculated as the average of $\bar{d}_k$ across all K units.

This can be applied to Pulse Asia senatorial pre-elections survey where we take $\hat{p}_k$ as the estimated proportion of voters who opted for candidate k and $n = 1,200$ as the common sample size. The higher C is, the less we are able to clearly distinguish between the proportion of votes received by different candidates, rendering ranks uninformative.

2.3.2 Pairwise Difference

Mohamad et al. (2019), requiring only the mean estimates and their corresponding standard errors similar to Klein et al. (2020), apply Tukey's Honest Significant Difference (HSD) to test $H_0 : \theta_j - \theta_k = 0$ for all $j \neq k \in 1, ..., K$ at level α. Tukey's HSD is typically used to provide simultaneous confidence statements about the differences between the means while controlling the family-wise error rate (FWER). Mohamad et al. (2019) come up with a joint confidence set for ranks expressed as

$$\left(1 + \#\left\{j : \frac{\hat{\theta}_j - \hat{\theta}_k}{\sqrt{\sigma_j^2 + \sigma_k^2}} > q_{1-\alpha}\right\},\ K - \#\left\{j : \frac{\hat{\theta}_j - \hat{\theta}_k}{\sqrt{\sigma_j^2 + \sigma_k^2}} < -q_{1-\alpha}\right\}\right), \tag{2.3.2}$$

$$\text{for } k = 1, 2, \ldots, K$$

where $q_{1-\alpha}$ is the $(1-\alpha)$ quantile of the distribution of the studentized range,

$$\max_{j,k=1,\ldots K} \frac{|\theta_j - \theta_k|}{\sqrt{\sigma_j^2 + \sigma_k^2}}.$$

In (2.3.2), the notation $\#\{\cdot\}$ counts the number of pairwise hypotheses that are rejected according to Tukey's HSD, which determines the lower and upper bounds of the confidence interval for the rank of $\hat{\theta}_k$. Their method has simultaneous coverage of at least $1 - \alpha$ and exactly $1 - \alpha$ when $\theta_1 = \theta_2 = \cdots = \theta_K$. However, their approach tends to be overly conservative,

showing coverage levels between 0.996 and 1.0 at a 0.90 nominal level in simulations, when the θs differ. They also demonstrated that as the true differences increase from 0 to 0.5, the coverage quickly increases from the nominal level to 1. As a remedy, they proposed a rescaling technique that brings the coverage closer to the nominal level, though it remains conservative (e.g., from 1.0 to 0.978, from 0.998 to 0.961—at 0.90 confidence level).

Mogstad et al. (2024) presents another technique that closely resembles the procedure by Klein et al. (2020) and Mohamad et al. (2019). However, they define ranks in the opposite way (i.e., larger rank value for lower estimate). They construct the rectangular confidence region in (2.3.3), from the pairwise differences of the estimators $\hat{\theta}_1, \ldots, \hat{\theta}_K$ and an estimator of the variance of $\hat{\theta}_j - \hat{\theta}_k$. The quantities $\hat{\theta}_1, \ldots, \hat{\theta}_K$ need not be independent. Let P_k be the distribution from which $\hat{\theta}_k$ is estimated; and let $\hat{P}_k$ denote the estimate of P_k. The joint confidence intervals for $r_1, \ldots, r_K$ are obtained from the joint confidence intervals of the pairwise difference:

$$C(1-\alpha, S) = \prod_{(j,k)\in S} \left[\hat{\theta}_j - \hat{\theta}_k \pm \sqrt{\sigma_j + \sigma_k}\, L^{-1}(1-\alpha, S, \hat{P})\right], \qquad S \subseteq \{(j,k) \in K \times K : j \neq k\} \tag{2.3.3}$$

where $L^{-1}(1-\alpha, S, P)$ is the $(1-\alpha)$-quantile corresponding to the CDF:

$$L(x, S, P) = P\left\{\max_{(j,k)\in S} \frac{|\hat{\theta}_j - \hat{\theta}_k - (\theta_j - \theta_k)|}{\sqrt{\sigma_j^2 + \sigma_k^2}} \leq x\right\}. \tag{2.3.4}$$

They added that if the estimators $\hat{\theta}_1, \hat{\theta}_2, \ldots, \hat{\theta}_K$ are jointly asymptotically normally distributed, then the quantiles $L^{-1}(1-\alpha, S, \hat{P})$, can be computed from the limiting distributions of the max-statistics shown in (2.3.4), through resampling methods. Their construction of confidence sets for ranks is based on hypothesis testing. Mogstad et al. (2024) show that their approach generally leads to a confidence set that is narrower than that of Klein et al. (2020).

2.3.3 Model-Based Uncertainty

In the context of ranking education and health institutions, Goldstein & Spiegelhalter (1996) treat the estimators as latent variables. Estimators are not directly measured but inferred from observed variables. They describe performance indicator as a summary statistical measurement on an institution (e.g., schools, hospitals, medical practitioners, etc.) which is meant to be related to the quality of its functioning. An example of performance indicator is "outcome" measure such as school exam results which have been used to judge institutional effectiveness. They use multilevel models to address the hierarchical nature of data structures associated with institutional performance. In its basic form,

$$
\begin{aligned}
y_{ik} &= \beta_0 + u_k + e_{ik} \\
\text{var}(\hat{u}_k) &= \sigma_u^2 \\
\text{var}(\hat{e}_{ik}) &= \sigma_e^2
\end{aligned}
$$

where y_{ik} is the exam score for student i from school k, u_k is the residual or the kth school effect, and e_{ik} is the residual effect for student i from school j. This will yield posterior estimates $\hat{u}_j$ and $\text{var}(\hat{u}_j)$ as well as $\text{rank}(\hat{u}_j)$ and $\text{var}(\text{rank}(\hat{u}_j))$ which can be used to compare the institutions. They also visualize rank uncertainty and describe that the non-overlap of confidence intervals convey a significant difference between compared institutions (Goldstein & Healy, 1995). Their example illustrates that while the multilevel model made individual estimates more accurate, it also had the effect of making the ranks even more uncertain.

2.3.4 Accounting for Data Dependencies

Some approaches explicitly account for dependencies in the data. Zhang et al. (2013) analyze U.S. age-adjusted cancer incidence and mortality rates

across states and counties by computing individual and overall simultaneous confidence intervals for age-adjusted health index using the Monte Carlo method. Because many health conditions are age-dependent, they use age-adjusted rates to minimize the confounding effect of age differences when comparing different population groups. They also extend their method to handle cases where only the adjusted rates and confidence intervals are available, aligning it more closely with the approach of Klein et al. (2020). Mohamad et al. (2019) show their technique to result in joint confidence sets with very low coverage probabilities and which are only able to reach the nominal level when differences among the means are large enough.

Hall & Miller (2009) mention that in some use cases such as institutions ranking, dependencies can be accommodated through conditioning, similar to the above approaches. However, in genomics where data on expression levels of different genes from the same individual are generally not independent, they suggest using an "independent component" version of the bootstrap on the sample, where m-out-of-n bootstrap (m $<$ n) is applied as though the ranked variables were statistically independent. They show this to perform at its best when a reasonable level of correlation is present among the variables.

Bazylik et al. (2025), in their recent study, tackle the ranking of political candidates or parties using the estimated share of support each one receives in surveys. They use the approach of Mogstad et al. (2024) but account for the dependence in the estimators through their bootstrap implementation. In their setup, they let X_k be the number of votes received by the kth party from the n voters so $\mathbf{X} = (X_1, X_2, \ldots, X_K)'$ is distributed according to the multinomial distribution with parameters n and $\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_K)'$. Their simulations show that bootstrap-based confidence sets may have coverage probability below the nominal level despite their being excessively wide. In contrast, the finite-sample confidence sets have coverage probability at least as large as expected and may even be relatively shorter.

2.4 Asymptotic Variance

Let $\hat{\theta}_{k,n}$ be an estimator of θ_k for $k = 1, 2, \ldots, K$ based on a sample of size n, and let $\hat{\theta}_{(k),n}$ be the kth ordered value among $\hat{\theta}_{1,n}, \hat{\theta}_{2,n}, \ldots, \hat{\theta}_{K,n}$. The estimation may be done through standard procedures such as MLE, least squares, and so forth. Moreover, suppose $\hat{\theta}_{k,n}$s are independent for $k = 1, 2, \ldots, K$. Finally, suppose, as in the application of this thesis, that $\hat{\theta}_{k,n}$ is a sample mean (or sample proportion) and θ_k is a population mean (or population proportion) and both $\hat{\theta}_{k,n}$ and θ_k are nonnegative. In this subsection, we show that

$$V[\hat{\theta}_{(k),n}] \longrightarrow k\text{th ordered value in } \{\theta_1^2 + \sigma_1^2, \theta_2^2 + \sigma_2^2, \ldots, \theta_K^2 + \sigma_K^2\} - \theta_{(k)}^2 \tag{2.4.1}$$

as $n \to \infty$, where $\sigma_k^2 = V(\hat{\theta}_{k,n})$, which in this study is assumed to be known. The result in (2.13) gives us an expression for the asymptotic variance of $\hat{\theta}_{(k),n}$. We shall also give an expression for an estimator of this variance.

By the definition of variance, we have

$$V(\hat{\theta}_{(k),n}) = E(\hat{\theta}_{(k),n}^2) - [E(\hat{\theta}_{(k),n})]^2 \tag{2.4.2}$$

We make use of Theorem 2.2 of Chen (1976) on asymptotic results for ordered estimators, which we shall now state.

Theorem (Chen, 1976). Suppose that the estimator $\hat{\delta}_{k,n}$ is a strongly consistent estimator of the parameter δ_k for $k = 1, 2, \ldots, K$ and based on a random sample of size n. Moreover, suppose $\hat{\delta}_{1,n}, \hat{\delta}_{2,n}, \ldots \hat{\delta}_{K,n}$ are independent. It follows that $\hat{\delta}_{(k),n}$ is a strongly consistent estimator of $\delta_{(k)}$, where $\hat{\delta}_{(k),n}$ is the kth ordered value of $\hat{\delta}_{1,n}, \hat{\delta}_{2,n}, \ldots, \hat{\delta}_{K,n}$ and $\delta_{(k)}$ is the kth ordered value of the $\delta_1, \delta_2, \ldots, \delta_K$.

Since $\hat{\theta}_{k,n}$, being a sample mean, is a strongly consistent estimator of θ_k, then using the result of Chen (1976), we know that $\hat{\theta}_{(k),n}$ is a strongly consistent estimator of $\theta_{(k)}$. This implies that

$$E[\hat{\theta}_{(k),n}] \longrightarrow \theta_{(k)}. \tag{2.4.3}$$

Moreover, since $E(\hat{\theta}^2_{k,n}) = \theta^2_k + \sigma^2_k$ we also have

$$E[\hat{\theta}^2_{(k),n}] \longrightarrow k\text{th ordered value in } \{\theta^2_1+\sigma^2_1, \theta^2_2+\sigma^2_2, \ldots, \theta^2_K+\sigma^2_K\}. \tag{2.4.4}$$

Using the results in (2.15) and (2.16) and noting the formula in (2.14), we get the result in (2.13). It also follows that we can estimate $V[\hat{\theta}_{(k),n}]$ as follows:

$$\widehat{V[\hat{\theta}_{(k),n}]} = \text{kth ordered value among } \left\{\hat{\theta}^2_{1,n} + \sigma^2_1, \ldots, \hat{\theta}^2_{K,n} + \sigma^2_K\right\} - \hat{\theta}^2_{(k),n}. \tag{2.4.5}$$

In this subsection, we have included the subscript n in our estimators to indicate that the convergence occurs as n tends to infinity. In the rest of this thesis, we shall drop this subscript.

3 Methodology

This section introduces the proposed methodologies to obtain joint confidence intervals that can be used to quantify uncertainty for the unknown overall true ranking. It addresses the scenario when the estimates of the parameters being ranked are correlated to certain degrees. Section 3.1 develops the procedure used to compute the joint confidence region for $\theta_1, \theta_2, \ldots, \theta_K$. These include a non-rank and rank-based methods. Section 3.2, on the other hand, develops the procedure to compute the joint confidence region for $\theta_{(1)}, \theta_{(2)}, \ldots, \theta_{(K)}$. Section 3.3 discusses some possible correlation structures to be considered for the estimates. Sections 3.4 and 3.5 discuss how to evaluate the proposed methodology and the simulation settings to be used.

3.1 Proposed methodology to compute the joint confidence region for the unordered parameters

In this section, we adopt the notations defined in Section 2.1 and 2.2. Moreover, we define $\hat{\boldsymbol{\theta}} = (\hat{\theta}_1, \hat{\theta}_2, \ldots, \hat{\theta}_K)'$ and assume that $\hat{\boldsymbol{\theta}} \sim N(\boldsymbol{\theta}, \boldsymbol{\Sigma})$ where $\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_K)'$ is unknown and $\boldsymbol{\Sigma}$ is a known $K \times K$ positive definite matrix. The diagonal elements of $\boldsymbol{\Sigma}$ are $\sigma_1^2, \ldots, \sigma_K^2$. We note that in the literature on inferences on the ranks, it is customary to assume that the variances are known. As previously mentioned, the setup in this thesis allows the estimates to be correlated, which is in contrast with that of Klein et al. (2020), which assumes independence among the estimates. In other words, in this thesis, the covariance matrix $\boldsymbol{\Sigma}$ need not be a diagonal matrix. Similar to the study of Klein et al. (2020), knowledge of the sampling design and estimation methodology for each population is not required; we shall first derive simultaneous confidence intervals for $\theta_1, \theta_2, \ldots, \theta_K$ of the form

$$\mathfrak{R}_1 = [\hat{\theta}_1 \pm t \times \sigma_1] \times [\hat{\theta}_2 \pm t \times \sigma_2] \times \cdots \times [\hat{\theta}_K \pm t \times \sigma_K]. \tag{3.1.1}$$

We want the joint confidence region in (3.1.1) to satisfy the following probability condition:

$$P\left(\hat{\theta}_k - t \cdot \sigma_k \leq \theta_k \leq \hat{\theta}_k + t \cdot \sigma_k,\ \forall\ k = 1, 2, \ldots, K\right) = 1 - \alpha. \tag{3.1.2}$$

Equivalently, we require

$$P\left(\max_{k=1,2,\ldots,K} \left|\frac{\hat{\theta}_k - \theta_k}{\sigma_k}\right| \leq t\right) = 1 - \alpha. \tag{3.1.3}$$

We use the parametric bootstrap to estimate the quantile t that will be used to construct the confidence intervals while controlling the coverage to be around the nominal level. In implementing the parametric bootstrap, we sample from the multivariate normal distribution with $\hat{\boldsymbol{\theta}}$ and $\boldsymbol{\Sigma}$ as mean and variance, respectively. The idea is also conceptually similar to that of Mogstad et al. (2024) who use resampling to obtain the quantile in a different context. Algorithm 1 describes the procedure to compute the joint confidence region based on unordered estimates.

Once the confidence intervals in (3.1.1) have been obtained, we can then use the result of Klein et al. (2020) in (2.2.4) to get the lower and upper bounds on the ranks $r_k, k = 1, 2, \ldots, K$. That is, we also get a joint confidence region for $r_1, r_2, ..., r_K$.

Algorithm 1 Computing the joint confidence region for the unordered parameters

Let the data be represented by $\hat{\boldsymbol{\theta}} = \left(\hat{\theta}_1, \hat{\theta}_2, \ldots, \hat{\theta}_K\right)'$ and suppose that $\boldsymbol{\Sigma}$ is known

1: **for** $b = 1, 2, \ldots, B$ **do**

2: Generate $\hat{\boldsymbol{\theta}}_b^* \sim N_K\left(\hat{\boldsymbol{\theta}}, \boldsymbol{\Sigma}\right)$ and write $\hat{\boldsymbol{\theta}}_b^* = \left(\hat{\theta}_{b1}^*, \hat{\theta}_{b2}^*, \ldots, \hat{\theta}_{bK}^*\right)'$

3: Compute

$$t_b^* = \max_{1 \le k \le K} \left| \frac{\hat{\theta}_{bk}^* - \hat{\theta}_k}{\sigma_k} \right|$$

4: **end for**

5: Compute the $(1-\alpha)$-sample quantile of $t_1^*, t_2^*, \ldots, t_B^*$, call this $\hat{t}$.

6: The joint confidence region for $\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_K)'$ is given by

$$\mathfrak{R}_1 = \left[\hat{\theta}_1 \pm \hat{t} \times \sigma_1\right] \times \left[\hat{\theta}_2 \pm \hat{t} \times \sigma_2\right] \times \cdots \times \left[\hat{\theta}_K \pm \hat{t} \times \sigma_K\right].$$

3.2 Proposed methodology to compute a joint confidence region for the ordered parameters

We shall also be proposing an approach to compute simultaneous confidence intervals for the ordered parameters. Suppose that for the parameters $\theta_1, \ldots, \theta_K$, the corresponding ordered values are $\theta_{(1)}, \ldots, \theta_{(K)}$. Similarly, let $\hat{\theta}_{(1)}, \ldots, \hat{\theta}_{(K)}$ be the ordered estimates. We now derive a joint confidence region for these ordered values, and set the region to be of the following form:

$$\mathfrak{R}_2 = [\hat{\theta}_{(1)} \pm t \times \sigma_{(1)}] \times [\hat{\theta}_{(2)} \pm t \times \sigma_{(2)}] \times \cdots \times [\hat{\theta}_{(K)} \pm t \times \sigma_{(K)}], \quad (3.2.1)$$

where $\sigma_{(k)} = \sqrt{V\left(\hat{\theta}_{(k)}\right)}$. The confidence region in (3.2.1) is computed such that the following condition holds:

$$P\left(\hat{\theta}_{(k)} - t \cdot \sigma_{(k)} \leq \theta_{(k)} \leq \hat{\theta}_{(k)} + t \cdot \sigma_{(k)},\ \forall k = 1, 2, \ldots, K\right) = 1 - \alpha, \tag{3.2.2}$$

which is equivalent to

$$P\left(\max_{k=1,2,\ldots,K} \left|\frac{\hat{\theta}_{(k)} - \theta_{(k)}}{\sigma_{(k)}}\right| \leq t\right) = 1 - \alpha, \tag{3.2.3}$$

and we estimate t, which in (3.2.3) is the $(1-\alpha)$-quantile of the distribution of $\max_k \left|\frac{\hat{\theta}_{(k)} - \theta_{(k)}}{\sigma_{(k)}}\right|$, via a suitable parametric bootstrap. Since $\sigma_{(k)}$ is not known, we estimate its value. We present two approaches to estimate $\sigma_{(k)}$. The first uses results from Chen (1976) and Dudewicz (1972) to obtain an expression of the asymptotic variance of $\hat{\theta}_{(k)}$ as follows:

$$V(\hat{\theta}_{(k)}) = \text{kth ordered value among } \left\{\theta_1^2 + \sigma_1^2, \theta_2^2 + \sigma_2^2, \ldots, \theta_K^2 + \sigma_K^2\right\} - \theta_{(k)}^2. \tag{3.2.4}$$

Consequently, we get an estimate of the asymptotic variance of $\hat{\theta}_{(k)}$, given by:

$$\widehat{V(\hat{\theta}_{(k)})} = \text{kth ordered value among } \left\{\hat{\theta}_1^2 + \sigma_1^2, \hat{\theta}_2^2 + \sigma_2^2, \ldots, \hat{\theta}_K^2 + \sigma_K^2\right\} - \hat{\theta}_{b(k)}^2. \tag{3.2.5}$$

Algorithm 2 describes the procedure to compute the joint confidence region in (3.2.1) using this approach.

Algorithm 2 Computing the joint confidence region for the ordered parameters using the estimated asymptotic variance

1: **for** $b = 1, 2, \ldots, B$ **do**

2: Generate $\hat{\boldsymbol{\theta}}_b^* = \left(\hat{\theta}_{b1}^*, \hat{\theta}_{b2}^*, \ldots, \hat{\theta}_{bK}^*\right)' \sim N_K\left(\hat{\boldsymbol{\theta}}, \boldsymbol{\Sigma}\right)$ and let $\hat{\theta}_{b(1)}^*, \hat{\theta}_{b(2)}^*, \ldots, \hat{\theta}_{b(K)}^*$ be the corresponding ordered values

3: Compute

$$\hat{\sigma}_{b(k)}^* = \sqrt{\left[\text{kth ordered value among } \left\{\hat{\theta}_{b1}^{*2} + \sigma_1^2, \hat{\theta}_{b2}^{*2} + \sigma_2^2, \ldots, \hat{\theta}_{bK}^{*2} + \sigma_K^2\right\}\right] - \hat{\theta}_{(k)}^{*2}}$$

4: Compute $t_b^* = \max\limits_{1 \le k \le K} \left| \frac{\hat{\theta}_{b(k)}^* - \hat{\theta}_{(k)}}{\hat{\sigma}_{b(k)}^*} \right|$

5: **end for**

6: Compute the $(1-\alpha)$-sample quantile of $t_1^*, t_2^*, \ldots, t_B^*$, call this $\hat{t}$.

7: The joint confidence region of $\theta_{(1)}, \theta_{(2)}, \ldots, \theta_{(K)}$ is given by

$$\mathfrak{R}_2 = \left[\hat{\theta}_{(1)} \pm \hat{t} \times \hat{\sigma}_{(1)}\right] \times \left[\hat{\theta}_{(2)} \pm \hat{t} \times \hat{\sigma}_{(2)}\right] \times \cdots \times \left[\hat{\theta}_{(K)} \pm \hat{t} \times \hat{\sigma}_{(K)}\right]$$

where $\hat{\sigma}_{(k)}$ is computed as

$$\hat{\sigma}_{(k)} = \sqrt{\text{kth ordered value among } \left\{\hat{\theta}_1^2 + \sigma_1^2, \hat{\theta}_2^2 + \sigma_2^2, \ldots, \hat{\theta}_K^2 + \sigma_K^2\right\} - \hat{\theta}_{(k)}^2}$$

The expression for the asymptotic variance of $\hat{\theta}_{(k)}$ has actually been derived in a setting where the $\hat{\theta}_1, \hat{\theta}_2, \ldots, \hat{\theta}_K$ are independent. It is not clear if the expression in (3.2.4) holds whenever $\hat{\theta}_1, \hat{\theta}_2, \ldots, \hat{\theta}_K$ are correlated. Since this thesis allows for the $\hat{\theta}_k$s to be correlated, it is possible that the expression in (3.2.5) for the estimate of the variance of $\hat{\theta}_{(k)}$ lacks optimality. As an alternative to using the asymptotic variance, we shall also consider using the bootstrap to estimate the variance of $\hat{\theta}_{(k)}$. To this end, a second-level bootstrap can be employed to estimate the variance. Algorithm 3 illustrates the procedure to compute the joint confidence region for the ordered parameters with the use of the bootstrap to estimate the variance of the ordered estimates. As may be expected, this approach is computationally intensive.

Algorithm 3 Computing the joint confidence region for the ordered parameters using the bootstrap estimate of the variance

1: **for** $b = 1, 2, \ldots, B$ **do**

2: Generate $\hat{\boldsymbol{\theta}}_b^* = \left(\hat{\theta}_{b1}^*, \hat{\theta}_{b2}^*, \ldots, \hat{\theta}_{bK}^*\right)' \sim N_K\left(\hat{\boldsymbol{\theta}}, \boldsymbol{\Sigma}\right)$ and let $\hat{\theta}_{b(1)}^*, \hat{\theta}_{b(2)}^*, \ldots, \hat{\theta}_{b(K)}^*$ be the corresponding ordered values of $\hat{\theta}_{b1}^*, \hat{\theta}_{b2}^*, \ldots, \hat{\theta}_{bK}^*$

3: **for** $c = 1, 2, \ldots, C$ **do**

4: Generate $\hat{\boldsymbol{\theta}}_{bc}^{**} = \left(\hat{\theta}_{bc1}^{**}, \hat{\theta}_{bc2}^{**}, \ldots, \hat{\theta}_{bcK}^{**}\right) \sim N_K\left(\hat{\boldsymbol{\theta}}_b^*, \boldsymbol{\Sigma}\right)$ and let $\hat{\theta}_{bc(1)}^{**}, \hat{\theta}_{bc(2)}^{**}, \ldots, \hat{\theta}_{bc(K)}^{**}$ be the corresponding ordered values of $\hat{\theta}_{bc1}^{**}, \hat{\theta}_{bc2}^{**}, \ldots, \hat{\theta}_{bcK}^{**}$

5: Compute $\hat{\sigma}_{b(k)}^* = \sqrt{\dfrac{\sum_{c=1}^{C}\left(\hat{\theta}_{bc(k)}^{**} - \bar{\hat{\theta}}_{b\cdot(k)}^{**}\right)^2}{C-1}}, \quad \bar{\hat{\theta}}_{b\cdot(k)}^{**} = \dfrac{1}{C}\sum_{c=1}^{C}\hat{\theta}_{bc(k)}^{**}$

6: **end for**

7: Compute $t_b^* = \max\limits_{1 \le k < K}\left|\frac{\hat{\theta}_{b(k)}^* - \hat{\theta}_{(k)}}{\hat{\sigma}_{b(k)}^*}\right|$

8: **end for**

9: Compute the $(1-\alpha)$-sample quantile of $t_1^*, t_2^*, \ldots, t_B^*$, call this $\hat{t}$.

10: The joint confidence region of $\theta_{(1)}, \theta_{(2)}, \ldots, \theta_{(K)}$ is

$$\mathfrak{R}_2 = \left[\hat{\theta}_{(1)} \pm \hat{t} \times \hat{\sigma}_{(1)}\right] \times \left[\hat{\theta}_{(2)} \pm \hat{t} \times \hat{\sigma}_{(2)}\right] \times \cdots \times \left[\hat{\theta}_{(K)} \pm \hat{t} \times \hat{\sigma}_{(K)}\right]$$

where $\hat{\sigma}_{(k)}$ is computed as

$$\hat{\sigma}_{(k)} = \sqrt{\frac{\sum_{b=1}^{B}\left(\hat{\theta}_{b(k)}^* - \bar{\hat{\theta}}_{\cdot(k)}^*\right)^2}{B-1}}, \quad \bar{\hat{\theta}}_{\cdot(k)}^* = \frac{1}{B}\sum_{b=1}^{B}\hat{\theta}_{b(k)}^*$$

3.3 Correlation structures

The proposed methodologies assume that $V(\hat{\boldsymbol{\theta}}) = \boldsymbol{\Sigma}$ is known. Note that we can express $\boldsymbol{\Sigma}$ as in (3.3.1), where $\mathbf{R}$ is the population correlation matrix.

$$\boldsymbol{\Sigma} = \boldsymbol{\Delta}^{1/2}\mathbf{R}\boldsymbol{\Delta}^{1/2}; \quad \boldsymbol{\Delta} = \operatorname{diag}\left\{\sigma_1^2, \sigma_2^2, \ldots, \sigma_K^2\right\}. \tag{3.3.1}$$

The diagonal elements of $\mathbf{\Sigma}$, which are $\sigma_k^2 = V(\hat{\theta}_k)$ for $k = 1, 2, \ldots, K$, are treated as known quantities in practice. With large sample sizes, the variance estimates are stable enough that they are treated as the actual values. However, there is limited information to use as basis for the correlations among the estimates. In this thesis, we shall be assuming certain correlation structures among the $\hat{\theta}$s.

One structure that may be used is an equicorrelation matrix shown in (3.3.2). This assumes that the K variables are equally correlated, i.e., that $\rho_{jk} = \rho$ where $\rho \in [-1, 1]$ for $j \neq k \in \{1, \ldots, K\}$.

$$\mathbf{R}_{\text{eq}} = (1-\rho)\,\mathbf{I}_K + \rho \mathbf{1}_K \mathbf{1}_K' = \begin{bmatrix} 1 & \rho & \cdots & \rho \\ \rho & 1 & \cdots & \rho \\ \vdots & \vdots & \ddots & \vdots \\ \rho & \rho & \cdots & 1 \end{bmatrix}_{K \times K} \tag{3.3.2}$$

In a block correlation matrix $\mathbf{R}_{block}$ with G blocks, as represented by Archakova & Hansen (2020), the correlation between any two variables is determined by the block to which the two variables belong. Each diagonal block represents an equicorrelation structure within group g, denoted by

$$\mathbf{R}_{\text{eq,g}} = (1-\rho_g)\,\mathbf{I}_{n_g} + \rho_g \mathbf{1}_{n_g} \mathbf{1}_{n_g}'$$

where ρ_g is the within-block correlation and n_g is the number of variables in block g such that $\Sigma_{g=1}^{G} n_g = K$. The off-diagonal blocks capture between-block correlations, represented by

$$\mathbf{C}_{g'g} = \mathbf{C}_{gg'} = \rho_{gg'} \mathbf{1}_{n_g} \mathbf{1}_{n_g}'$$
$$\text{where } g \neq g' \in \{1, \ldots, G\}$$

Thus, the full block correlation matrix can be expressed as in (3.3.3).

$$\mathbf{R}_{\text{block}} = \begin{bmatrix} \mathbf{R}_{eq,1} & \mathbf{C}_{12} & \cdots & \mathbf{C}_{1G} \\ \mathbf{C}_{21} & \mathbf{R}_{eq,2} & \cdots & \mathbf{C}_{2G} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{C}_{G1} & \mathbf{C}_{G2} & \cdots & \mathbf{R}_{eq,G} \end{bmatrix}_{K \times K} \tag{3.3.3}$$

In the context of pre-election surveys, each block may represent correlations induced by party or ticket membership, reflecting stronger associations within parties and weaker associations between them.

Correlation structures that account for spatial proximity can be borrowed from geostatistics. This is particularly relevant in light of Klein's observation that states located within certain regions exhibit similar travel time characteristics. In such cases, spatial dependence can be modeled using a stationary (i.e., no directional dependence) Matérn correlation function, which for two locations $\mathbf{s}_i$ and $\mathbf{s}_j$ is expressed as in (3.3.4).

$$\rho_{\text{matern}} = \frac{2^{1-\nu}}{\Gamma(\nu)}(\kappa \parallel \mathbf{s}_i - \mathbf{s}_j \parallel)^{\nu} K_{\nu}(\kappa \parallel \mathbf{s}_i - \mathbf{s}_j \parallel) \tag{3.3.4}$$

where $\| \cdot \|$ denotes the Euclidean distance and K_ν is the second kind of the modified Bessel function. It has a scale parameter $\kappa > 0$ and a smoothness parameter $\nu > 0$. ρ_{matern} reduces to the exponential correlation when $\nu = 0.5$ and to Gaussian correlation function when $\nu = \infty$. In this paper, the R package "BayesNGSP" (Turek & Risser (2022)), is used to construct the $\mathbf{R}_{\text{matern}}$.

3.4 Evaluation

Algorithm 4 estimates the coverage probabilities associated with the proposed methodologies. The coverage probabilities correspond to the proportion of replications in which the true parameter values are contained within the confidence intervals for all K simultaneously. Moreover, the tightness of the joint confidence region that results from Algorithm 1 is assessed using three summary measures: the arithmetic mean (T_1), geometric mean (T_2), and the metric T_3 introduced by Wright (2025), as presented in Equations 3.4.1–3.4.3.

$$T_1 = \frac{1}{K} \sum_{k=1}^{K} \left| \Lambda_{Ok} \right| \tag{3.4.1}$$

$$T_2 = \left[\prod_{k=1}^{K} \left| \Lambda_{Ok} \right| \right]^{1/K} \tag{3.4.2}$$

$$T_3 = 1 - \frac{OP}{K^2} \tag{3.4.3}$$

In equation 3.4.3, $OP = K + \sum_{k=1}^{K} \left|\Lambda_{Ok}\right|$ denotes the total number of occupied positions in a joint confidence region out of the total number of positions K^2; or the sum of the differences between the upper and lower bound of the simultaneous rank intervals added by 1, for each k. Higher values of T_1 and T_2 indicate wider confidence intervals and are therefore less desirable, whereas higher values of T_3 are preferable. T_3 can range from 0, indicating no tightness, to $\frac{K-1}{K}$, implying the confidence region only contains the estimated ranking which is likely the true ranking.

Algorithm 4 Computing the coverage probability and tightness measures

For given values of $\boldsymbol{\Sigma}$ and $\theta_1, \theta_2, \ldots, \theta_K$ (with corresponding $\theta_{(1)}, \theta_{(2)}, \ldots, \theta_{(K)}$ for rank-based methods)

1: **for** replications $= 1, 2, \ldots, 5000$ **do**

2: Generate $\hat{\boldsymbol{\theta}} \sim N_K(\boldsymbol{\theta}, \boldsymbol{\Sigma})$

3: Compute the confidence region for the unordered (ordered) parameters using Algorithm 1 ($\mathfrak{R}_1$) (Algorithms 2 and 3 ($\mathfrak{R}_2$)), Bonferroni ($\mathfrak{R}_{\text{bonf}}$), and independence assumption ($\mathfrak{R}_{\text{ind}}$).

4: Check if $(\theta_1, \theta_2, \ldots, \theta_K) \in \mathfrak{R}_1, \mathfrak{R}_{\text{bonf}}, \mathfrak{R}_{\text{ind}}$ $\left(\left(\theta_{(1)}, \theta_{(2)}, \ldots, \theta_{(K)}\right) \in \mathfrak{R}_2\right)$ for the unordered parameters (ordered parameters) and compute T_1, T_2, and T_3.

5: **end for**

6: Compute the coverage in line 4 is satisfied and the average of T_1, T_2, and T_3. Results of Algorithm 1 ($\mathfrak{R}_1$) is compared to ($\mathfrak{R}_{\text{bonf}}$ and $\mathfrak{R}_{\text{ind}}$).

3.5 Simulation settings

For the simulation settings to be used in evaluating the performance of the proposed methodologies, we shall be varying the settings for the population mean $\boldsymbol{\theta}$, population variances $\sigma_1^2, \sigma_2^2, \ldots, \sigma_K^2$, population correlation matrix $\mathbf{R}$, and number of populations being ranked K. In carrying out the simulations, a nominal level of $1 - \alpha = 0.95$ will be used. Table 3.5.1 summarizes the simulation settings.

Table 3.5.1: True $\boldsymbol{\theta}$ values

K	Variance	$\boldsymbol{\theta}$	$\mathbf{R}$
10	High	19.9, 27, 34.9, 25.3, 26.7, 21.9, 21.3, 17.3, 27.7, 9.4	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
	Low	22.5, 24.9, 27.5, 24.3, 24.8, 23.2, 23, 21.6, 25.1, 19	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
	Moderate	21.5, 25.7, 30.5, 24.7, 25.5, 22.6, 22.3, 19.9, 26.1, 15.2	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
20	High	19.9, 27, 34.9, 25.3, 26.7, 21.9, 21.3, 17.3, 27.7, 9.4, 16.2, 24.6, 31.9, 33, 21.9, 29.9, 17.7, 17.3, 29.3, 16	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
	Low	22.5, 24.9, 27.5, 24.3, 24.8, 23.2, 23, 21.6, 25.1, 19, 21.3, 24.1, 26.5, 26.9, 23.2, 25.8, 21.8, 21.6, 25.6, 21.2	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
	Moderate	21.5, 25.7, 30.5, 24.7, 25.5, 22.6, 22.3, 19.9, 26.1, 15.2, 19.3, 24.3, 28.7, 29.3, 22.7, 27.5, 20.1, 19.9, 27.1, 19.1	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
30	High	19.9, 27, 34.9, 25.3, 26.7, 21.9, 21.3, 17.3, 27.7, 9.4, 16.2, 24.6, 31.9, 33, 21.9, 29.9, 17.7, 17.3, 29.3, 16, 29.4, 19, 20.4, 20.3, 18.2, 25, 23.3, 18.3, 26, 27.4	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
	Low	22.5, 24.9, 27.5, 24.3, 24.8, 23.2, 23, 21.6, 25.1, 19, 21.3, 24.1, 26.5, 26.9, 23.2, 25.8, 21.8, 21.6, 25.6, 21.2, 25.7, 22.2, 22.7, 22.6, 21.9, 24.2, 23.6, 22, 24.5, 25	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
	Moderate	21.5, 25.7, 30.5, 24.7, 25.5, 22.6, 22.3, 19.9, 26.1, 15.2, 19.3, 24.3, 28.7, 29.3, 22.7, 27.5, 20.1, 19.9, 27.1, 19.1, 27.2, 20.9, 21.8, 21.7, 20.4, 24.5, 23.5, 20.5, 25.1, 26	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks

Table 3.5.1: True $\boldsymbol{\theta}$ values *(continued)*

K	Variance	$\boldsymbol{\theta}$	$\mathbf{R}$
40	High	19.9, 27, 34.9, 25.3, 26.7, 21.9, 21.3, 17.3, 27.7, 9.4, 16.2, 24.6, 31.9, 33, 21.9, 29.9, 17.7, 17.3, 29.3, 16, 29.4, 19, 20.4, 20.3, 18.2, 25, 23.3, 18.3, 26, 27.4, 31.9, 27.7, 28.1, 33.6, 28.2, 20.9, 14.1, 15.6, 30.5, 30.5	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
	Low	22.5, 24.9, 27.5, 24.3, 24.8, 23.2, 23, 21.6, 25.1, 19, 21.3, 24.1, 26.5, 26.9, 23.2, 25.8, 21.8, 21.6, 25.6, 21.2, 25.7, 22.2, 22.7, 22.6, 21.9, 24.2, 23.6, 22, 24.5, 25, 26.5, 25.1, 25.2, 27.1, 25.3, 22.8, 20.6, 21.1, 26, 26	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
	Moderate	21.5, 25.7, 30.5, 24.7, 25.5, 22.6, 22.3, 19.9, 26.1, 15.2, 19.3, 24.3, 28.7, 29.3, 22.7, 27.5, 20.1, 19.9, 27.1, 19.1, 27.2, 20.9, 21.8, 21.7, 20.4, 24.5, 23.5, 20.5, 25.1, 26, 28.6, 26.1, 26.4, 29.7, 26.5, 22.1, 18, 18.9, 27.8, 27.8	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
50	High	19.9, 27, 34.9, 25.3, 26.7, 21.9, 21.3, 17.3, 27.7, 9.4, 16.2, 24.6, 31.9, 33, 21.9, 29.9, 17.7, 17.3, 29.3, 16, 29.4, 19, 20.4, 20.3, 18.2, 25, 23.3, 18.3, 26, 27.4, 31.9, 27.7, 28.1, 33.6, 28.2, 20.9, 14.1, 15.6, 30.5, 30.5, 18.8, 16.6, 22.1, 19.4, 19.6, 21.6, 16.7, 26.7, 33.5, 22.5	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
	Low	22.5, 24.9, 27.5, 24.3, 24.8, 23.2, 23, 21.6, 25.1, 19, 21.3, 24.1, 26.5, 26.9, 23.2, 25.8, 21.8, 21.6, 25.6, 21.2, 25.7, 22.2, 22.7, 22.6, 21.9, 24.2, 23.6, 22, 24.5, 25, 26.5, 25.1, 25.2, 27.1, 25.3, 22.8, 20.6, 21.1, 26, 26, 22.1, 21.4, 23.2, 22.3, 22.4, 23.1, 21.4, 24.8, 27, 23.4	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
	Moderate	21.5, 25.7, 30.5, 24.7, 25.5, 22.6, 22.3, 19.9, 26.1, 15.2, 19.3, 24.3, 28.7, 29.3, 22.7, 27.5, 20.1, 19.9, 27.1, 19.1, 27.2, 20.9, 21.8, 21.7, 20.4, 24.5, 23.5, 20.5, 25.1, 26, 28.6, 26.1, 26.4, 29.7, 26.5, 22.1, 18, 18.9, 27.8, 27.8, 20.8, 19.5, 22.8, 21.2, 21.3, 22.5, 19.5, 25.5, 29.6, 23	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks

The following configurations are used for $\mathbf{R}_{block}$:

Table 3.5.2: Configuration for $\mathbf{R}_{block}$

2 Blocks				3 Blocks			
$\mathbf{R}_{block}$	Block No.	Size	ρ	$\mathbf{R}_{block}$	Block No.	Size	ρ
B2	1	0.5K x 0.5K	0.9	UL3	1	0.2K x 0.2K	0.9
	2	0.5K x 0.5K	0.1		2	0.3K x 0.3K	0.5
					3	0.5K x 0.5K	0.1
U2	1	0.3K x 0.3K	0.9	UH3	1	0.2K x 0.2K	0.1
	2	0.7K x 0.7K	0.1		2	0.3K x 0.3K	0.5
					3	0.5K x 0.5K	0.9

Table 3.5.3: σ Values.

K	σ
10	0.14, 0.33, 0.15, 0.23, 0.07, 0.19, 0.19, 0.37, 0.32, 0.11
20	0.14, 0.33, 0.15, 0.23, 0.07, 0.19, 0.19, 0.37, 0.32, 0.11, 0.17, 0.27, 0.24, 0.11, 0.11, 0.13, 0.16, 0.15, 0.15, 0.25
30	0.14, 0.33, 0.15, 0.23, 0.07, 0.19, 0.19, 0.37, 0.32, 0.11, 0.17, 0.27, 0.24, 0.11, 0.11, 0.13, 0.16, 0.15, 0.15, 0.25, 0.15, 0.13, 0.1, 0.1, 0.24, 0.13, 0.32, 0.19, 0.27, 0.3
40	0.14, 0.33, 0.15, 0.23, 0.07, 0.19, 0.19, 0.37, 0.32, 0.11, 0.17, 0.27, 0.24, 0.11, 0.11, 0.13, 0.16, 0.15, 0.15, 0.25, 0.15, 0.13, 0.1, 0.1, 0.24, 0.13, 0.32, 0.19, 0.27, 0.3, 0.12, 0.27, 0.09, 0.12, 0.36, 0.09, 0.15, 0.16, 0.09, 0.29
50	0.14, 0.33, 0.15, 0.23, 0.07, 0.19, 0.19, 0.37, 0.32, 0.11, 0.17, 0.27, 0.24, 0.11, 0.11, 0.13, 0.16, 0.15, 0.15, 0.25, 0.15, 0.13, 0.1, 0.1, 0.24, 0.13, 0.32, 0.19, 0.27, 0.3, 0.12, 0.27, 0.09, 0.12, 0.36, 0.09, 0.15, 0.16, 0.09, 0.29, 0.16, 0.28, 0.14, 0.07, 0.2, 0.31, 0.13, 0.14, 0.31, 0.11

4 Result

The outcomes of the simulation studies are organized into Sections 4.1 and 4.2 for the unordered and ordered parameters, respectively.

4.1 Joint confidence region for the unordered parameters

The resulting joint confidence regions for the unordered parameters obtained through Algorithm 1 are compared to those from Klein's independent assumption and Bonferroni approaches. This is to show the advantage of using the said algorithm in the presence of correlation among ranked estimates.

The first two facets in Figure 4.1.1 illustrates that coverage values for both KWW-Ind and KWW-Bonf joint confidence regions increase with higher ρ in an equicorrelated structure, irrespective of K, reflecting the conservative nature of these regions. Meanwhile, Algorithm 1, shown in the third facet, consistently produces coverage that is close to the nominal level, 0.95. The advantage in using the approach is evident at moderate to high ρ.

In the case of block diagonal correlation structures presented in the second row of Figure 4.1.1, the improvement from using Algorithm 1 becomes distinguishable when larger block sizes have higher ρ. This is particularly evident for B2 and UH3 where 50% of the estimates are set to be highly correlated. In U2, 30% of the estimates are highly correlated, while in UL3, 20% are highly correlated. In both cases, the proposed algorithm has little to no advantage over independent and Bonferroni.

The improvement in using Algorithm 1 is diminished for block diagonal correlation structure compared to that of equicorrelation structure. This is because the correlation among estimates is weaker in the block diagonal

structure, where off-diagonal blocks are zero.

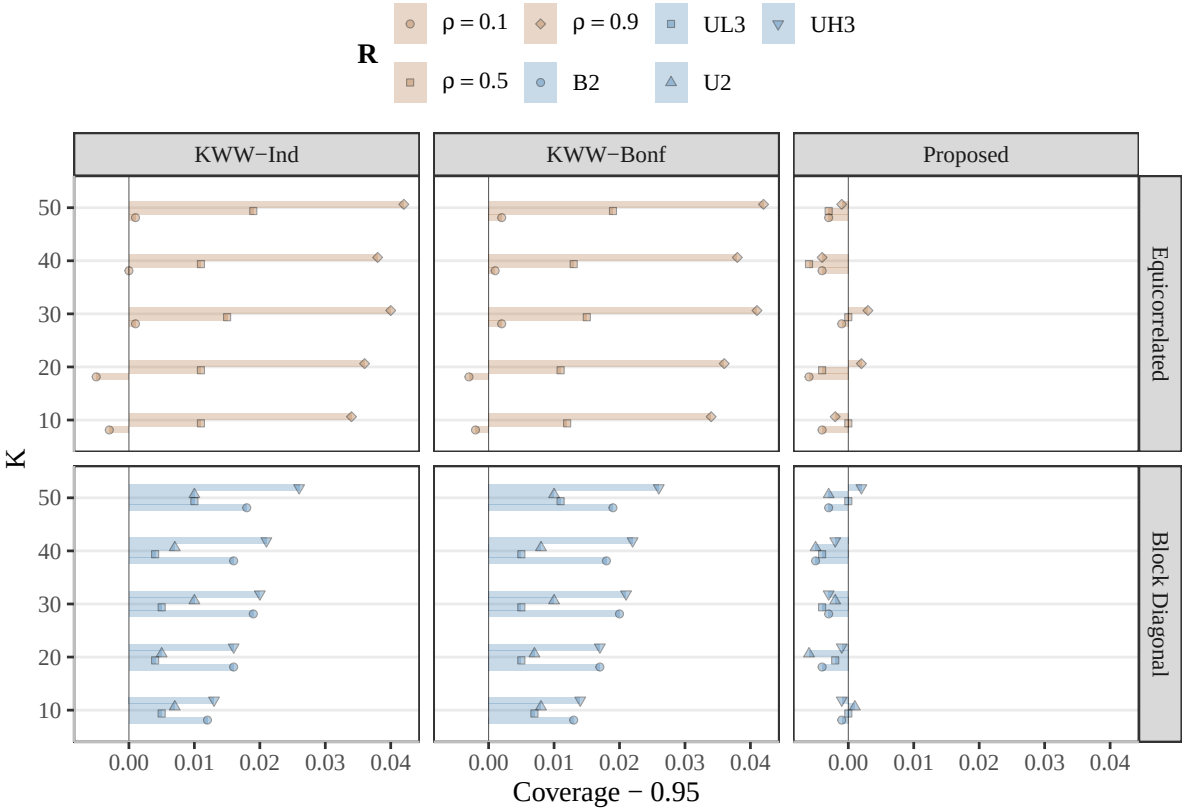

Figure 4.1.1: Simulation Result for Coverage of Unordered Parameters

The same coverage values are observed for different spreads of the true parameter vector $\boldsymbol{\theta}$. This results from assuming the same known Σ across cases. In Equation (3.1.3), it follows that the distribution of t only depends on Σ and K. This is more intuitive in the bootstrap world as demonstrated in Algorithm 1. In step 2, we generate $\hat{\boldsymbol{\theta}}_b^* \sim N_K(\hat{\boldsymbol{\theta}}, \Sigma)$ and compute $\max_k \left| \frac{\hat{\theta}_{bk}^* - \hat{\theta}_k}{\sigma_k} \right|$ where $\hat{\boldsymbol{\theta}}_b^* - \hat{\boldsymbol{\theta}} \sim N_K(0, \Sigma)$. From Equation (3.3.1), this also implies that the correlation matrix $\mathbf{R}$ influences the coverage. The coverage values corresponding to Figure 4.1.1 are provided in Table A.1 in the Appendix.

Results for T_1 and T_2 tightness measures for $\mathbf{R}_{\text{eq}}$, shown in rows 1 and 2 of Figure 4.1.2, indicate that Algorithm 1 yields narrower intervals than both the independent and Bonferroni approaches. The improvement is negligible at low ρ and becomes more pronounced as ρ increases; in each facet of Figure 4.1.2, this is captured by the bend as each line connecting the points goes from KWW-Ind, KWW-Bonf, to Proposed. The angle of the bend in each

facet also shows that the improvement in T_1 and T_2 is larger with lower variance in the estimates. Moreover, while intervals become wider with larger K, as shown by the groupwise movement of the points from the left to the right facet for each row of Figure 4.1.2, this effect is dampened by the variance of the estimates. That is, the general increase in T_1 and T_2 as K increases is larger for estimates that vary less. This is demonstrated by the corresponding jumps of each group of same-size points across facets for every row. When $\hat{\theta}$ values are more dispersed, the intervals are narrower; however, as they vary less or become closer to each other, ranks tend to interchange more, resulting in wider ranges. This aligns with the intuition that intervals are most precise when the estimates being ranked have greater separation. The same observations can be made for T_3 noting that higher values are more desirable. Altogether, these trends show that the proposed approach is most advantageous in the presence of moderate to high correlation, especially when ranking a larger number of estimates with close values.

The last three rows of Figure 4.1.2 show the results for a block diagonal correlation structure. Similar to the coverage results, the overall benefit of using the Proposed is reduced compared to when an equicorrelated structure is enforced. In each facet, this is illustrated through the muted bend of the line connecting the points corresponding to KWW-Bonf and the Proposed. Otherwise, the patterns observed for the equicorrelated structure remain consistent.

The actual values corresponding to T_1, T_2 and T_2 in Figure 4.1.2 are provided in Tables A.2, A.3, and A.4 of the Appendix.

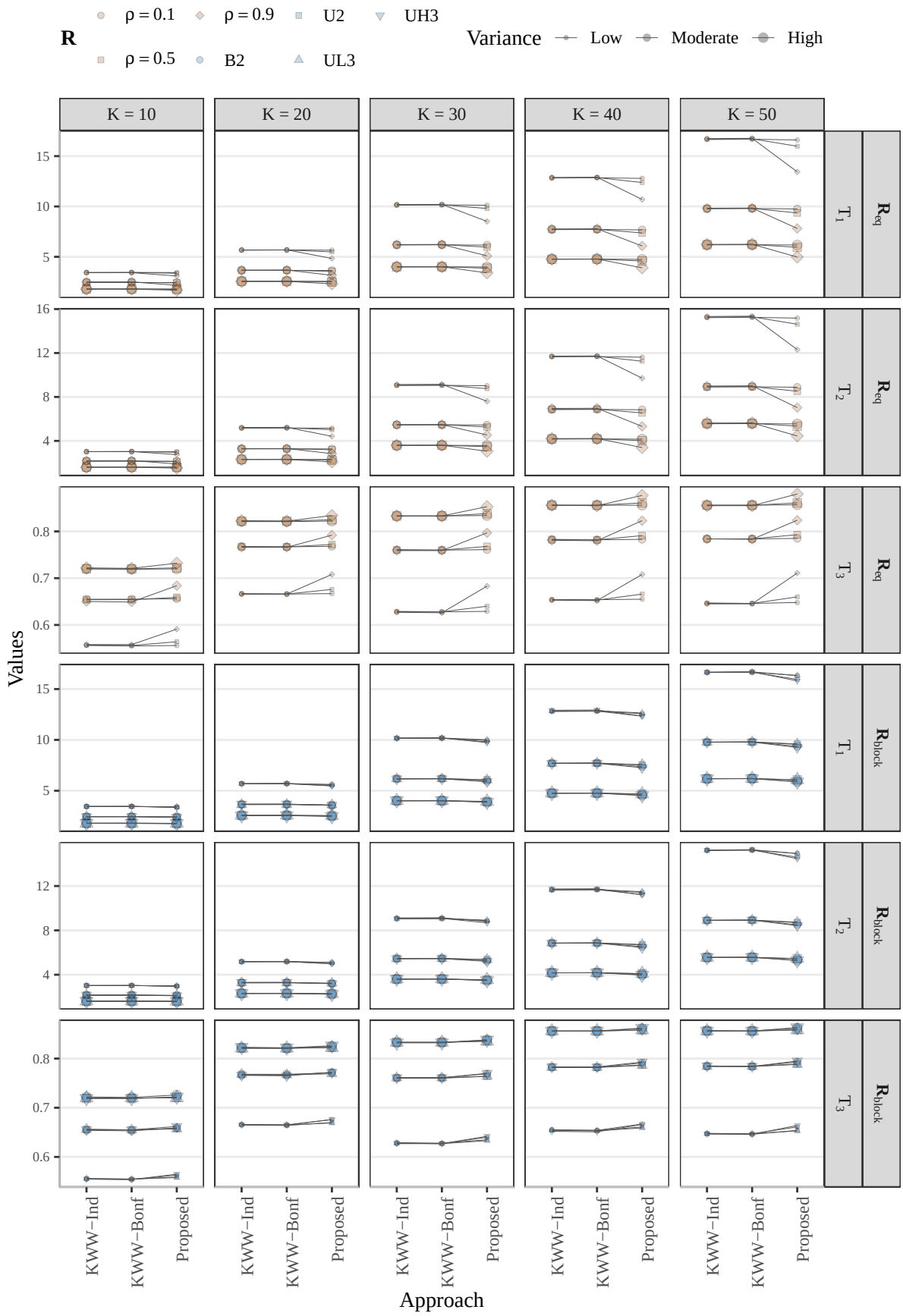

Figure 4.1.2: Simulation Result for Tightness Measure of Unordered Parameters

4.2 Joint confidence region for the ordered parameters

Figure 4.2.1 illustrates results for joint confidence region for the ordered parameters. With confidence regions for ranks, we are not able to map the intervals to the estimates, unlike those in the previous section.

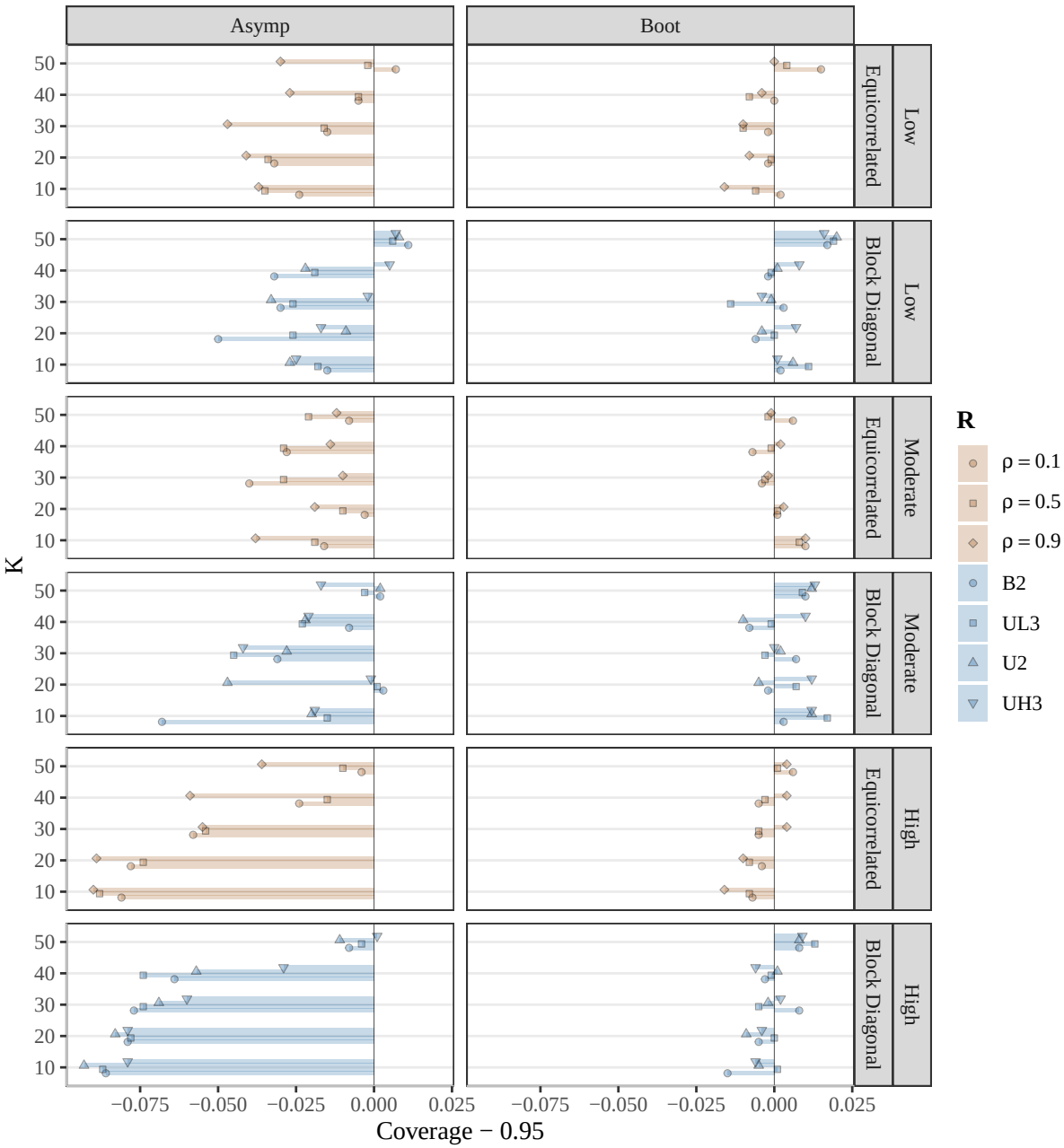

Figure 4.2.1: Simulation Result for Coverage of Ordered Parameters

With the two proposed approaches, Asymp is seen to have coverage less than the nominal level while Boot produced values close to 0.95 except for

$K = 50$. For this case, it Boot tends to be conservative.

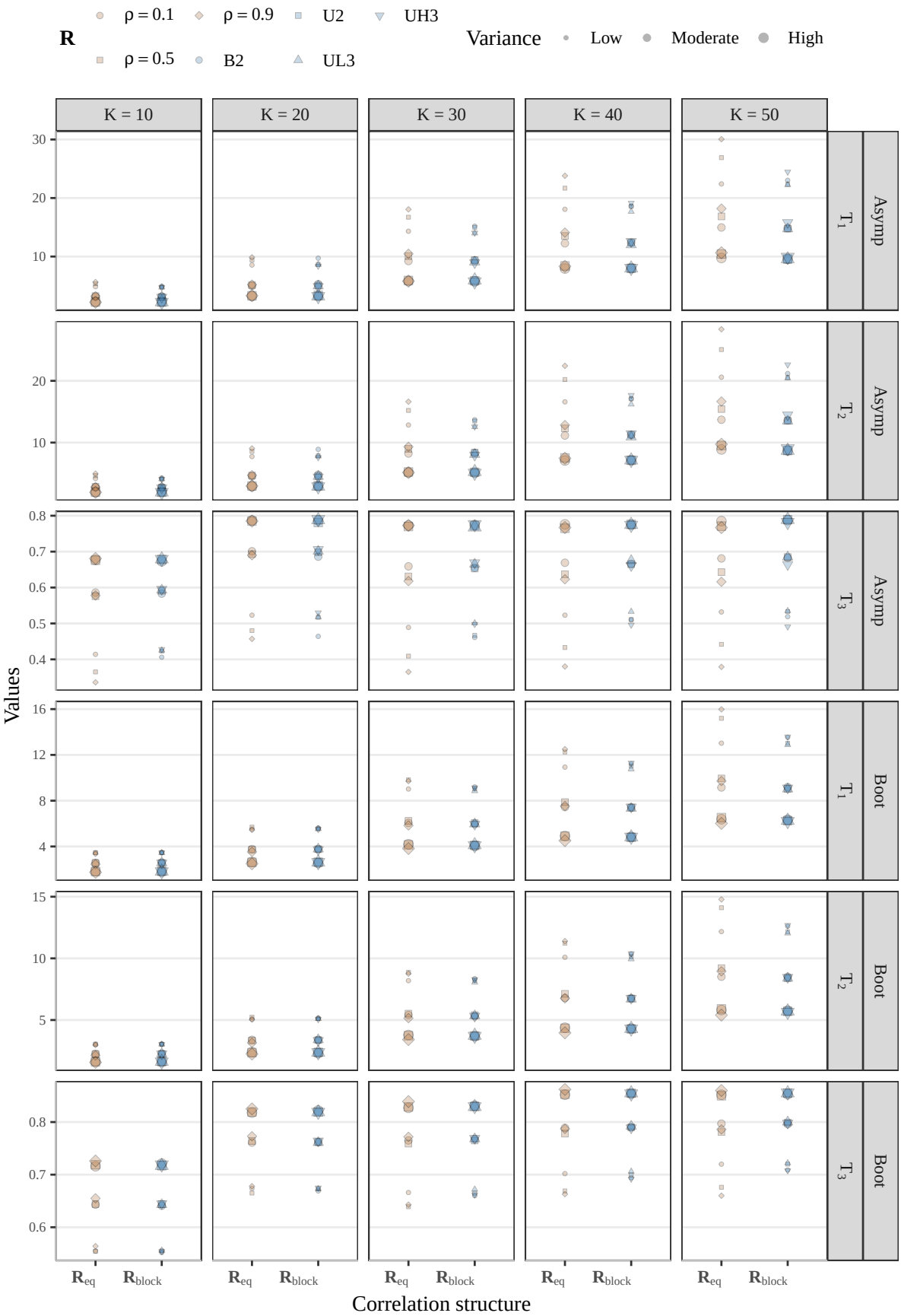

Figure 4.2.2: Simulation Result for Tightness Measure of Ordered Parameters

5 Application

5.1 Pulse Asia Results

For the Pulse Asia results, correlations are assumed to be induced by alliances or political slates formed for the purpose of elections. Slates campaigned by the incumbent president (Alyansa) and his former allies, the vice-president and former president (DuterTen), are considered. Imee Marcos and Villar are both endorsed by these rival factions and are given a separate block. Makabayan is also given its own block as it has a relatively large set of 11 candidates. Another popular alliance that emerged is that of Pangilinan and Aquino (KiBam). The remaining candidates are collapsed into a single block.

Table 5.1.1: Joint Confidence Region Tightness Measures

ρ	Approach	T_1	T_2	T_3
0.0	KWW-Ind	19.08	15.386	0.598
	KWW-Bonf	19.20	15.543	0.596
	Proposed	19.20	15.543	0.596
0.1		18.84	15.207	0.603
0.2		19.08	15.386	0.598
0.3		18.82	15.198	0.604
0.4		18.64	14.965	0.607
0.5		18.64	14.965	0.607
0.6		18.32	14.625	0.614
0.7		18.32	14.625	0.614
0.8		17.92	14.189	0.622
0.9		17.28	13.539	0.634

Table 5.1.1 shows that higher within-block correlation leads to better T1, T2, and T3. This means that overall, the intervals in the joint confidence region are generally narrower. Figure 5.1.1 assumes high correlations within blocks of alliances and political slates. The joint confidence regions for

Independent, Bonferroni, and Nonrank are shown respectively. For this use case, lower rank corresponds to highly preferred candidates.

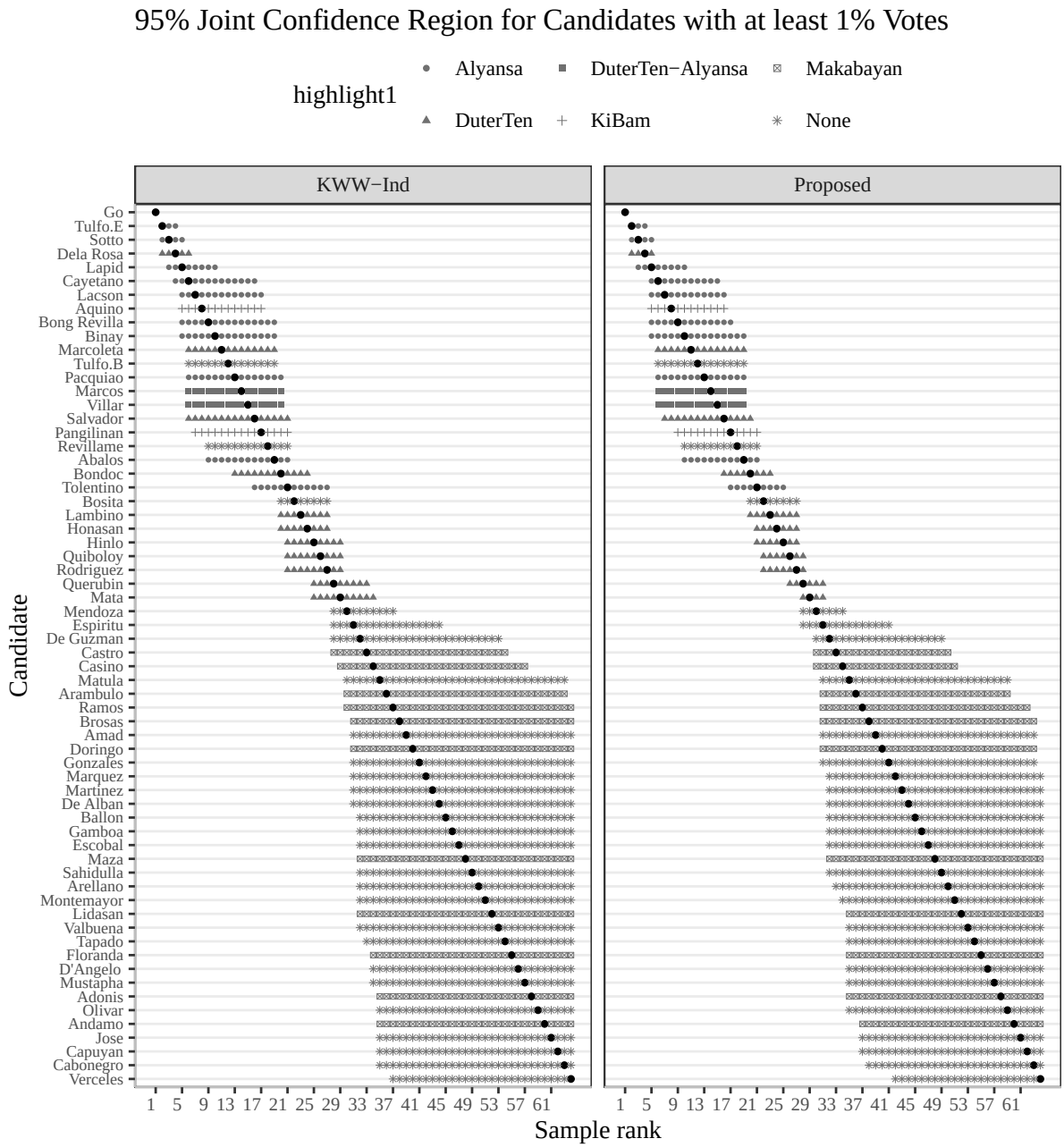

Figure 5.1.1: 95% Joint Confidence Region for Candidate Votes

5.2 2011 Mean Travel Time to Work in the US

Block diagonal correlation structure is assumed according to 2010 population density in each corresponding state (https://www.energy.gov/cmei/vehicles/fact-661-february-7-2011-population-density). Five blocks are used wherein the first has population density from 0 to 20 people per square mile,

the second has 20 to 50, the third has 50 to 100, the fourth has 100 to 200 and the fifth has more than 200.

Table 5.2.1: Joint Confidence Region Tightness Measures

ρ	Approach	T_1	T_2	T_3
0.0	KWW-Ind	10.86	9.244	0.763
	KWW-Bonf	10.86	9.244	0.763
	Proposed	10.82	9.210	0.764
0.1		10.82	9.210	0.764
0.2		10.98	9.318	0.760
0.3		10.74	9.143	0.765
0.4		10.74	9.143	0.765
0.5		10.74	9.143	0.765
0.6		10.46	8.927	0.771
0.7		10.38	8.881	0.772
0.8		9.98	8.503	0.780
0.9		9.70	8.083	0.786

Figure 5.2.1 assumes high correlations within blocks with different population densities. Only joint confidence regions for Bonferroni and Nonrank are since Independent result is similar to Bonferroni. For this use case, lower rank corresponds to states with lower mean travel time.

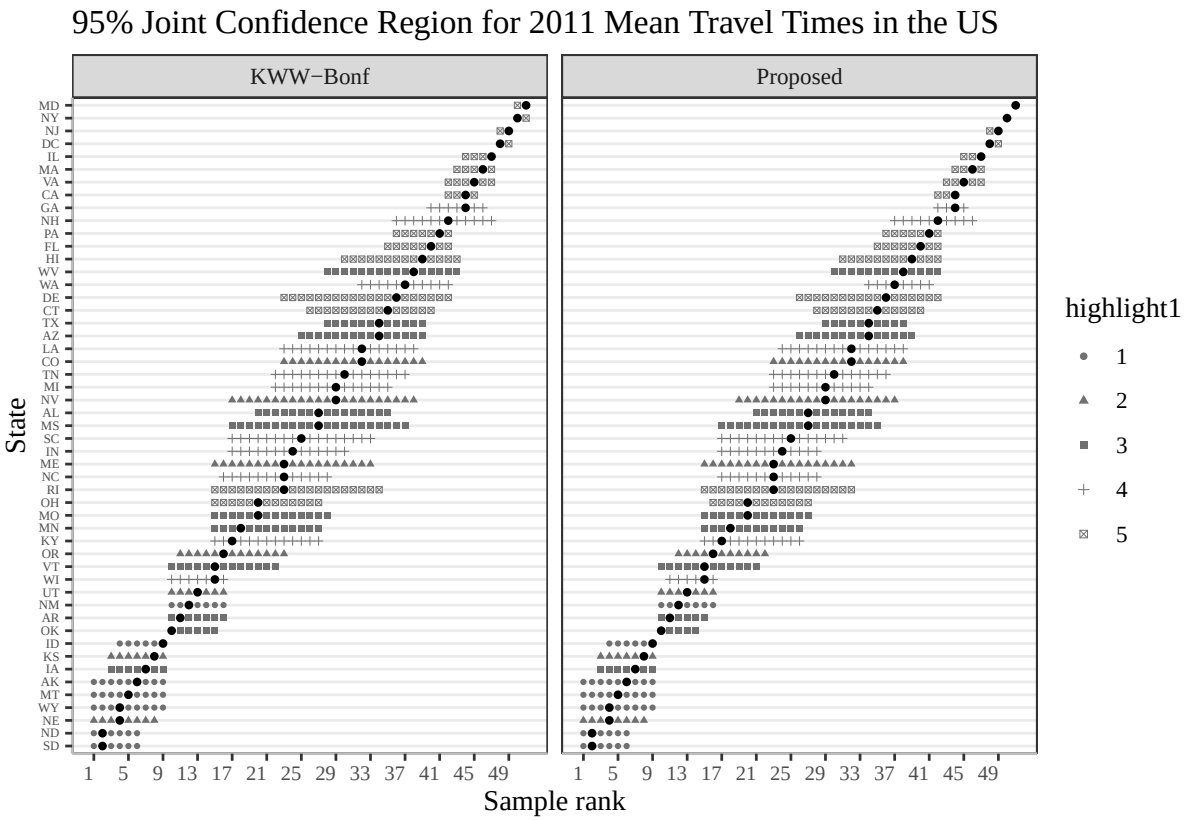

Figure 5.2.1: 95% Joint Confidence Region for 2011 Mean Travel Times in the US

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Appendix

Figure A.1 shows how the spread of the θ varies in the cases considered in Table 3.5.1

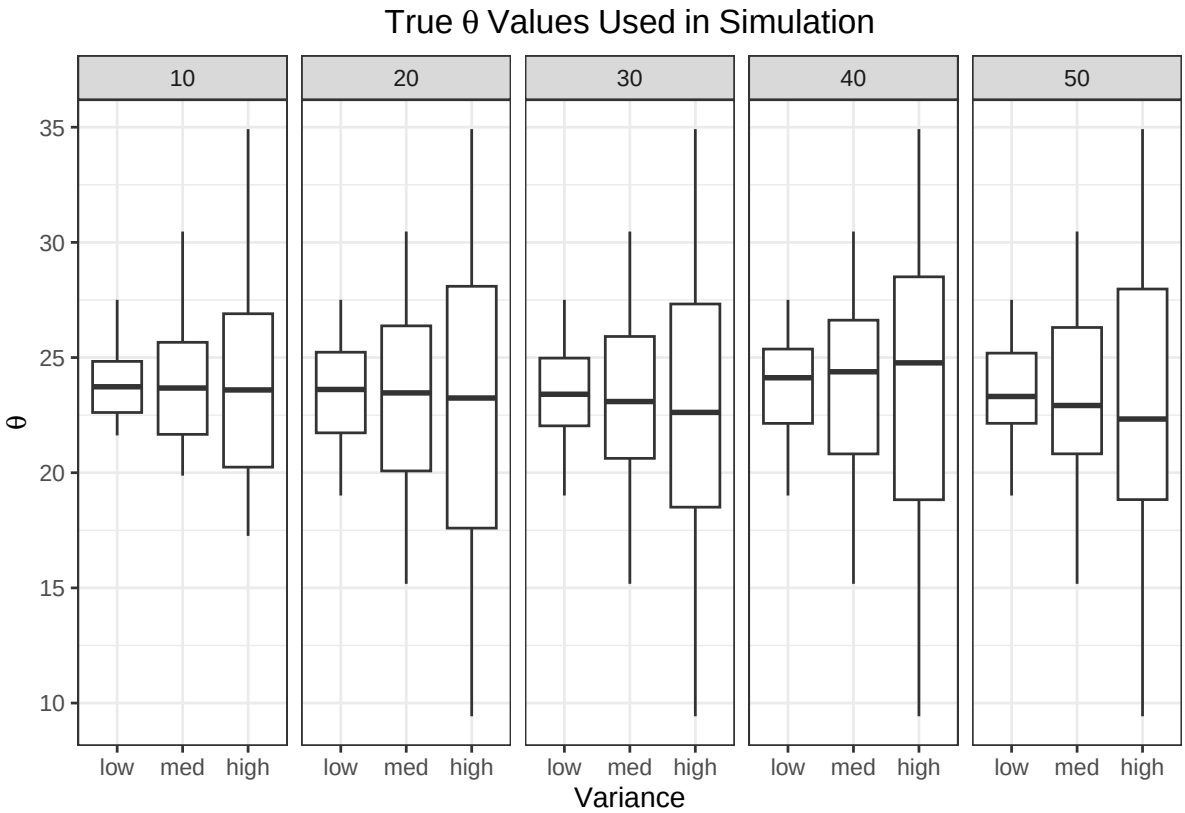

Figure A.1: True θ values used in the simulation have different levels of variability.

Table A.1: Simulation Results for Coverage of Unordered Parameters

Equicorrelated				Block Diagonal			
ρ	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$		$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$
K = 10							
0.1	0.947	0.948	0.946	B2	0.962	0.963	0.949
0.5	0.961	0.962	0.950	U2	0.957	0.958	0.951
0.9	0.984	0.984	0.948	UL3	0.955	0.957	0.950
				UH3	0.963	0.964	0.949
K = 20							
0.1	0.945	0.947	0.944	B2	0.966	0.967	0.946
0.5	0.961	0.961	0.946	U2	0.955	0.957	0.944
0.9	0.986	0.986	0.952	UL3	0.954	0.955	0.948
				UH3	0.966	0.967	0.949
K = 30							
0.1	0.951	0.952	0.949	B2	0.969	0.970	0.947
0.5	0.965	0.965	0.950	U2	0.960	0.960	0.948
0.9	0.990	0.991	0.953	UL3	0.955	0.955	0.946
				UH3	0.970	0.971	0.947
K = 40							
0.1	0.950	0.951	0.946	B2	0.966	0.968	0.945
0.5	0.961	0.963	0.944	U2	0.957	0.958	0.945
0.9	0.988	0.988	0.946	UL3	0.954	0.955	0.946
				UH3	0.971	0.972	0.948
K = 50							
0.1	0.951	0.952	0.947	B2	0.968	0.969	0.947
0.5	0.969	0.969	0.947	U2	0.960	0.960	0.947
0.9	0.992	0.992	0.949	UL3	0.960	0.961	0.950
				UH3	0.976	0.976	0.952

Table A.2: Simulation Results for T_1 of Unordered Parameters

		Equicorrelated									Block Diagonal									
		Low			Moderate			High				Low			Moderate			High		
K	ρ	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$		$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$
10	0.1	3.444	3.451	3.436	2.446	2.451	2.440	1.803	1.805	1.802	B2	3.455	3.461	3.375	2.464	2.469	2.406	1.784	1.787	1.745
	0.5	3.434	3.440	3.363	2.460	2.464	2.405	1.800	1.802	1.775	U2	3.444	3.450	3.406	2.442	2.447	2.416	1.806	1.808	1.794
	0.9	3.416	3.423	3.094	2.504	2.508	2.165	1.785	1.787	1.677	UL3	3.441	3.448	3.417	2.445	2.449	2.429	1.802	1.804	1.795
											UH3	3.454	3.460	3.351	2.446	2.451	2.379	1.809	1.811	1.780
20	0.1	5.673	5.684	5.661	3.645	3.652	3.638	2.572	2.575	2.568	B2	5.682	5.694	5.474	3.674	3.680	3.561	2.569	2.573	2.497
	0.5	5.664	5.675	5.489	3.659	3.666	3.556	2.562	2.566	2.508	U2	5.700	5.711	5.605	3.637	3.642	3.584	2.573	2.578	2.539
	0.9	5.677	5.688	4.838	3.680	3.686	3.163	2.549	2.553	2.319	UL3	5.682	5.693	5.611	3.649	3.655	3.610	2.579	2.584	2.553
											UH3	5.695	5.706	5.484	3.685	3.692	3.560	2.550	2.554	2.472
30	0.1	10.142	10.162	10.116	6.181	6.194	6.162	4.000	4.007	3.989	B2	10.179	10.197	9.804	6.171	6.183	5.933	4.011	4.019	3.868
	0.5	10.148	10.167	9.797	6.194	6.207	5.959	4.002	4.009	3.868	U2	10.142	10.161	9.950	6.193	6.205	6.075	4.001	4.008	3.929
	0.9	10.188	10.207	8.519	6.216	6.228	5.091	4.006	4.013	3.402	UL3	10.168	10.188	10.012	6.165	6.177	6.068	4.015	4.022	3.955
											UH3	10.175	10.193	9.734	6.162	6.174	5.888	4.032	4.039	3.870
40	0.1	12.824	12.848	12.789	7.700	7.716	7.676	4.760	4.770	4.746	B2	12.804	12.827	12.331	7.705	7.721	7.368	4.754	4.763	4.562
	0.5	12.845	12.869	12.378	7.716	7.733	7.369	4.765	4.773	4.580	U2	12.809	12.831	12.573	7.698	7.715	7.532	4.758	4.767	4.668
	0.9	12.882	12.902	10.697	7.767	7.784	6.076	4.775	4.784	3.902	UL3	12.845	12.868	12.645	7.704	7.720	7.563	4.760	4.768	4.681
											UH3	12.918	12.940	12.341	7.676	7.692	7.260	4.751	4.761	4.525
50	0.1	16.649	16.682	16.602	9.780	9.800	9.754	6.188	6.200	6.172	B2	16.623	16.653	15.950	9.781	9.799	9.366	6.171	6.184	5.905
	0.5	16.685	16.718	15.981	9.788	9.807	9.358	6.206	6.218	5.943	U2	16.659	16.690	16.315	9.771	9.791	9.559	6.187	6.199	6.061
	0.9	16.738	16.771	13.437	9.824	9.843	7.814	6.232	6.244	4.976	UL3	16.634	16.666	16.337	9.783	9.803	9.602	6.185	6.196	6.075
											UH3	16.654	16.687	15.806	9.765	9.782	9.254	6.186	6.198	5.872

Table A.3: Simulation Results for T_2 of Unordered Parameters

		Equicorrelated									Block Diagonal									
		Low			Moderate			High				Low			Moderate			High		
K	ρ	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$		$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$
10	0.1	3.018	3.023	3.011	2.150	2.155	2.146	1.612	1.614	1.612	B2	3.025	3.029	2.961	2.168	2.171	2.117	1.598	1.600	1.568
	0.5	3.020	3.025	2.964	2.160	2.164	2.112	1.611	1.612	1.593	U2	3.018	3.022	2.988	2.149	2.153	2.125	1.615	1.616	1.606
	0.9	3.021	3.026	2.752	2.196	2.200	1.903	1.599	1.601	1.521	UL3	3.014	3.020	2.995	2.149	2.153	2.135	1.612	1.614	1.607
											UH3	3.031	3.036	2.947	2.156	2.161	2.097	1.619	1.621	1.598
20	0.1	5.163	5.174	5.154	3.273	3.279	3.266	2.313	2.316	2.309	B2	5.174	5.185	4.988	3.300	3.306	3.199	2.308	2.312	2.245
	0.5	5.176	5.186	5.018	3.282	3.288	3.190	2.312	2.315	2.261	U2	5.187	5.197	5.104	3.264	3.269	3.216	2.315	2.319	2.284
	0.9	5.222	5.232	4.417	3.298	3.303	2.842	2.309	2.313	2.088	UL3	5.172	5.182	5.109	3.280	3.285	3.244	2.320	2.324	2.297
											UH3	5.179	5.188	4.989	3.302	3.309	3.188	2.312	2.315	2.240
30	0.1	9.041	9.059	9.018	5.458	5.470	5.442	3.588	3.595	3.579	B2	9.125	9.141	8.802	5.445	5.454	5.244	3.612	3.619	3.476
	0.5	9.065	9.082	8.754	5.463	5.474	5.261	3.596	3.603	3.473	U2	9.038	9.055	8.868	5.468	5.479	5.365	3.590	3.597	3.525
	0.9	9.134	9.150	7.609	5.472	5.482	4.538	3.600	3.607	3.055	UL3	9.058	9.075	8.920	5.450	5.460	5.367	3.605	3.611	3.550
											UH3	9.066	9.082	8.675	5.444	5.454	5.207	3.620	3.627	3.475
40	0.1	11.653	11.675	11.620	6.853	6.869	6.831	4.171	4.179	4.159	B2	11.660	11.681	11.225	6.849	6.864	6.537	4.172	4.180	3.999
	0.5	11.681	11.704	11.250	6.873	6.889	6.545	4.183	4.190	4.017	U2	11.634	11.655	11.415	6.844	6.859	6.689	4.171	4.180	4.091
	0.9	11.717	11.735	9.707	6.956	6.972	5.318	4.210	4.217	3.395	UL3	11.666	11.688	11.481	6.859	6.874	6.727	4.173	4.180	4.102
											UH3	11.735	11.755	11.200	6.837	6.852	6.446	4.169	4.178	3.968
50	0.1	15.218	15.248	15.174	8.906	8.925	8.882	5.559	5.570	5.544	B2	15.211	15.239	14.598	8.903	8.920	8.513	5.552	5.563	5.308
	0.5	15.264	15.294	14.618	8.925	8.943	8.518	5.584	5.595	5.342	U2	15.244	15.272	14.930	8.895	8.913	8.697	5.562	5.572	5.446
	0.9	15.325	15.355	12.307	8.993	9.012	7.029	5.624	5.635	4.447	UL3	15.204	15.233	14.931	8.911	8.929	8.741	5.558	5.568	5.458
											UH3	15.239	15.270	14.462	8.906	8.923	8.423	5.559	5.570	5.275

Table A.4: Simulation Results for T_3 of Unordered Parameters

		Equicorrelated									Block Diagonal									
		Low			Moderate			High				Low			Moderate			High		
K	ρ	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$		$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$
10	0.1	0.556	0.555	0.556	0.655	0.655	0.656	0.720	0.719	0.720	B2	0.555	0.554	0.563	0.654	0.653	0.659	0.722	0.721	0.726
	0.5	0.557	0.556	0.564	0.654	0.654	0.659	0.720	0.720	0.722	U2	0.556	0.555	0.559	0.656	0.655	0.658	0.719	0.719	0.721
	0.9	0.558	0.558	0.591	0.650	0.649	0.684	0.722	0.721	0.732	UL3	0.556	0.555	0.558	0.656	0.655	0.657	0.720	0.720	0.720
											UH3	0.555	0.554	0.565	0.655	0.655	0.662	0.719	0.719	0.722
20	0.1	0.666	0.666	0.667	0.768	0.767	0.768	0.821	0.821	0.822	B2	0.666	0.665	0.676	0.766	0.766	0.772	0.822	0.821	0.825
	0.5	0.667	0.666	0.676	0.767	0.767	0.772	0.822	0.822	0.825	U2	0.665	0.664	0.670	0.768	0.768	0.771	0.821	0.821	0.823
	0.9	0.666	0.666	0.708	0.766	0.766	0.792	0.823	0.822	0.834	UL3	0.666	0.665	0.669	0.768	0.767	0.769	0.821	0.821	0.822
											UH3	0.665	0.665	0.676	0.766	0.765	0.772	0.823	0.822	0.826
30	0.1	0.629	0.628	0.629	0.761	0.760	0.761	0.833	0.833	0.834	B2	0.627	0.627	0.640	0.761	0.761	0.769	0.833	0.833	0.838
	0.5	0.628	0.628	0.640	0.760	0.760	0.768	0.833	0.833	0.838	U2	0.629	0.628	0.635	0.760	0.760	0.764	0.833	0.833	0.836
	0.9	0.627	0.626	0.683	0.759	0.759	0.797	0.833	0.833	0.853	UL3	0.628	0.627	0.633	0.761	0.761	0.764	0.833	0.833	0.835
											UH3	0.628	0.627	0.642	0.761	0.761	0.770	0.832	0.832	0.838
40	0.1	0.654	0.654	0.655	0.783	0.782	0.783	0.856	0.856	0.856	B2	0.655	0.654	0.667	0.782	0.782	0.791	0.856	0.856	0.861
	0.5	0.654	0.653	0.666	0.782	0.782	0.791	0.856	0.856	0.861	U2	0.655	0.654	0.661	0.783	0.782	0.787	0.856	0.856	0.858
	0.9	0.653	0.652	0.708	0.781	0.780	0.823	0.856	0.855	0.877	UL3	0.654	0.653	0.659	0.782	0.782	0.786	0.856	0.856	0.858
											UH3	0.652	0.651	0.666	0.783	0.783	0.793	0.856	0.856	0.862
50	0.1	0.647	0.646	0.648	0.784	0.784	0.785	0.856	0.856	0.857	B2	0.648	0.647	0.661	0.784	0.784	0.793	0.857	0.856	0.862
	0.5	0.646	0.646	0.660	0.784	0.784	0.793	0.856	0.856	0.861	U2	0.647	0.646	0.654	0.785	0.784	0.789	0.856	0.856	0.859
	0.9	0.645	0.645	0.711	0.784	0.783	0.824	0.855	0.855	0.880	UL3	0.647	0.647	0.653	0.784	0.784	0.788	0.856	0.856	0.858
											UH3	0.647	0.646	0.664	0.785	0.784	0.795	0.856	0.856	0.863

Table A.5: Simulation Results for Coverage of Ordered Parameters

Equicorrelated			Block Diagonal		
ρ	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$		$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$
K = 10					
0.1	0.926	0.952	B2	0.935	0.952
0.5	0.915	0.944	U2	0.923	0.956
0.9	0.913	0.934	UL3	0.932	0.961
			UH3	0.925	0.951
K = 20					
0.1	0.918	0.948	B2	0.900	0.944
0.5	0.916	0.949	U2	0.941	0.946
0.9	0.909	0.942	UL3	0.924	0.950
			UH3	0.933	0.957
K = 30					
0.1	0.935	0.948	B2	0.920	0.953
0.5	0.934	0.940	U2	0.917	0.949
0.9	0.903	0.940	UL3	0.924	0.936
			UH3	0.948	0.946
K = 40					
0.1	0.945	0.950	B2	0.918	0.948
0.5	0.945	0.942	U2	0.928	0.951
0.9	0.923	0.946	UL3	0.931	0.949
			UH3	0.955	0.958
K = 50					
0.1	0.957	0.965	B2	0.961	0.967
0.5	0.948	0.954	U2	0.958	0.970
0.9	0.920	0.950	UL3	0.956	0.969
			UH3	0.957	0.966

Table A.6: Simulation Results for Coverage of Ordered Parameters

Equicorrelated			Block Diagonal		
ρ	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$		$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$
K = 10					
0.1	0.934	0.960	B2	0.882	0.953
0.5	0.931	0.958	U2	0.930	0.962
0.9	0.912	0.960	UL3	0.935	0.967
			UH3	0.931	0.962
K = 20					
0.1	0.947	0.951	B2	0.953	0.948
0.5	0.940	0.951	U2	0.903	0.945
0.9	0.931	0.953	UL3	0.951	0.957
			UH3	0.949	0.962
K = 30					
0.1	0.910	0.946	B2	0.919	0.957
0.5	0.921	0.947	U2	0.922	0.952
0.9	0.940	0.948	UL3	0.905	0.947
			UH3	0.908	0.950
K = 40					
0.1	0.922	0.943	B2	0.942	0.942
0.5	0.921	0.949	U2	0.928	0.940
0.9	0.936	0.952	UL3	0.927	0.949
			UH3	0.929	0.960
K = 50					
0.1	0.942	0.956	B2	0.952	0.960
0.5	0.929	0.948	U2	0.952	0.962
0.9	0.938	0.949	UL3	0.947	0.959
			UH3	0.933	0.963

Table A.7: Simulation Results for Coverage of Ordered Parameters

Equicorrelated			Block Diagonal		
ρ	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$		$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$
K = 10					
0.1	0.869	0.943	B2	0.864	0.935
0.5	0.862	0.942	U2	0.857	0.945
0.9	0.860	0.934	UL3	0.863	0.951
			UH3	0.871	0.944
K = 20					
0.1	0.872	0.946	B2	0.871	0.945
0.5	0.876	0.942	U2	0.867	0.941
0.9	0.861	0.940	UL3	0.872	0.950
			UH3	0.871	0.946
K = 30					
0.1	0.892	0.945	B2	0.873	0.958
0.5	0.896	0.945	U2	0.881	0.948
0.9	0.895	0.954	UL3	0.876	0.945
			UH3	0.890	0.952
K = 40					
0.1	0.926	0.945	B2	0.886	0.947
0.5	0.935	0.947	U2	0.893	0.951
0.9	0.891	0.954	UL3	0.876	0.949
			UH3	0.921	0.944
K = 50					
0.1	0.946	0.956	B2	0.942	0.958
0.5	0.940	0.951	U2	0.939	0.958
0.9	0.914	0.954	UL3	0.946	0.963
			UH3	0.951	0.959

Table A.8: Simulation Results for T_1 of Ordered Parameters

		Equicorrelated						Block Diagonal						
		Low		Moderate		High			Low		Moderate		High	
K	ρ	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$		$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$
10	0.1	4.859	3.453	3.137	2.574	2.228	1.841	B2	4.935	3.483	3.168	2.591	2.268	1.793
	0.5	5.352	3.460	3.236	2.559	2.235	1.809	U2	4.766	3.454	3.092	2.561	2.218	1.831
	0.9	5.638	3.357	3.222	2.447	2.191	1.736	UL3	4.752	3.444	3.074	2.570	2.213	1.828
								UH3	4.719	3.461	3.047	2.553	2.211	1.808
20	0.1	8.546	5.517	4.984	3.772	3.258	2.647	B2	9.718	5.623	5.285	3.769	3.265	2.557
	0.5	9.407	5.691	5.131	3.727	3.297	2.615	U2	8.655	5.546	5.042	3.759	3.371	2.628
	0.9	9.852	5.450	5.201	3.533	3.307	2.493	UL3	8.631	5.540	4.977	3.775	3.233	2.614
								UH3	8.391	5.499	4.876	3.719	3.203	2.617
30	0.1	14.319	9.019	9.226	6.058	5.811	4.193	B2	15.183	9.200	9.341	5.972	5.772	4.059
	0.5	16.719	9.818	10.077	6.239	5.909	4.142	U2	14.978	9.041	9.407	6.051	5.849	4.097
	0.9	18.060	9.711	10.471	5.837	5.799	3.820	UL3	13.977	8.851	9.058	5.965	5.950	4.127
								UH3	14.046	9.138	8.935	5.942	5.716	4.093
40	0.1	18.076	10.931	12.259	7.443	7.950	4.911	B2	18.614	11.223	12.369	7.404	7.997	4.834
	0.5	21.678	12.251	13.511	7.874	8.396	4.894	U2	18.580	11.019	12.383	7.404	7.980	4.800
	0.9	23.799	12.496	14.093	7.538	8.332	4.513	UL3	17.680	10.754	12.004	7.322	7.992	4.811
								UH3	19.168	11.317	12.580	7.448	8.037	4.833
50	0.1	22.405	13.024	14.966	9.155	9.753	6.381	B2	23.029	13.533	14.860	9.184	9.587	6.331
	0.5	26.895	15.196	16.858	9.943	10.435	6.522	U2	22.341	13.011	14.761	9.031	9.560	6.257
	0.9	30.046	15.978	18.194	9.717	10.642	6.015	UL3	22.236	12.877	14.733	9.022	9.554	6.263
								UH3	24.466	13.587	15.807	9.106	9.913	6.204

Table A.9: Simulation Results for T_2 of Ordered Parameters

		Equicorrelated						Block Diagonal						
		Low		Moderate		High			Low		Moderate		High	
K	ρ	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$		$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$
10	0.1	4.188	3.035	2.754	2.260	1.973	1.636	B2	4.270	3.055	2.776	2.271	2.016	1.601
	0.5	4.652	3.043	2.834	2.238	1.983	1.612	U2	4.107	3.037	2.716	2.250	1.965	1.629
	0.9	4.968	2.969	2.810	2.126	1.953	1.559	UL3	4.102	3.028	2.702	2.255	1.960	1.627
								UH3	4.071	3.037	2.679	2.248	1.961	1.614
20	0.1	7.738	5.083	4.530	3.391	2.927	2.390	B2	8.910	5.164	4.797	3.384	2.936	2.313
	0.5	8.577	5.236	4.666	3.336	2.965	2.356	U2	7.846	5.107	4.581	3.375	3.030	2.378
	0.9	9.053	5.030	4.715	3.158	2.979	2.241	UL3	7.812	5.102	4.523	3.398	2.904	2.368
								UH3	7.596	5.055	4.425	3.343	2.885	2.374
30	0.1	12.843	8.191	8.232	5.408	5.152	3.784	B2	13.697	8.355	8.329	5.312	5.121	3.684
	0.5	15.211	8.861	8.993	5.518	5.238	3.729	U2	13.485	8.209	8.387	5.395	5.185	3.705
	0.9	16.623	8.763	9.312	5.152	5.149	3.415	UL3	12.522	8.053	8.079	5.342	5.272	3.738
								UH3	12.576	8.289	7.961	5.313	5.074	3.700
40	0.1	16.595	10.089	11.144	6.794	7.098	4.375	B2	17.110	10.365	11.255	6.751	7.146	4.306
	0.5	20.211	11.226	12.299	7.124	7.520	4.326	U2	17.083	10.144	11.255	6.748	7.127	4.276
	0.9	22.428	11.398	12.834	6.758	7.464	3.962	UL3	16.208	9.941	10.904	6.698	7.142	4.292
								UH3	17.669	10.394	11.439	6.762	7.179	4.295
50	0.1	20.590	12.169	13.701	8.514	8.892	5.810	B2	21.180	12.601	13.617	8.501	8.736	5.758
	0.5	25.067	14.107	15.436	9.206	9.537	5.899	U2	20.524	12.155	13.527	8.390	8.711	5.701
	0.9	28.350	14.781	16.655	8.951	9.729	5.399	UL3	20.416	12.037	13.490	8.391	8.706	5.709
								UH3	22.622	12.688	14.477	8.460	9.049	5.645

Table A.10: Simulation Results for T_3 of Ordered Parameters

		Equicorrelated						Block Diagonal						
		Low		Moderate		High			Low		Moderate		High	
K	ρ	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$		$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$
10	0.1	0.414	0.555	0.586	0.643	0.677	0.716	B2	0.406	0.552	0.583	0.641	0.673	0.721
	0.5	0.365	0.554	0.576	0.644	0.676	0.719	U2	0.423	0.555	0.591	0.644	0.678	0.717
	0.9	0.336	0.564	0.578	0.655	0.681	0.726	UL3	0.425	0.556	0.593	0.643	0.679	0.717
								UH3	0.428	0.554	0.595	0.645	0.679	0.719
20	0.1	0.523	0.674	0.701	0.761	0.787	0.818	B2	0.464	0.669	0.686	0.762	0.787	0.822
	0.5	0.480	0.665	0.693	0.764	0.785	0.819	U2	0.517	0.673	0.698	0.762	0.781	0.819
	0.9	0.457	0.678	0.690	0.773	0.785	0.825	UL3	0.518	0.673	0.701	0.761	0.788	0.819
								UH3	0.530	0.675	0.706	0.764	0.790	0.819
30	0.1	0.489	0.666	0.659	0.765	0.773	0.827	B2	0.461	0.660	0.655	0.768	0.774	0.831
	0.5	0.409	0.639	0.631	0.759	0.770	0.829	U2	0.467	0.665	0.653	0.765	0.772	0.830
	0.9	0.365	0.643	0.618	0.772	0.773	0.839	UL3	0.501	0.672	0.665	0.768	0.768	0.829
								UH3	0.498	0.662	0.669	0.769	0.776	0.830
40	0.1	0.523	0.702	0.669	0.789	0.776	0.852	B2	0.510	0.694	0.666	0.790	0.775	0.854
	0.5	0.433	0.669	0.637	0.778	0.765	0.853	U2	0.511	0.700	0.665	0.790	0.776	0.855
	0.9	0.380	0.663	0.623	0.787	0.767	0.862	UL3	0.533	0.706	0.675	0.792	0.775	0.855
								UH3	0.496	0.692	0.661	0.789	0.774	0.854
50	0.1	0.532	0.720	0.681	0.797	0.785	0.852	B2	0.519	0.709	0.683	0.796	0.788	0.853
	0.5	0.442	0.676	0.643	0.781	0.771	0.850	U2	0.533	0.720	0.685	0.799	0.789	0.855
	0.9	0.379	0.660	0.616	0.786	0.767	0.860	UL3	0.535	0.722	0.685	0.800	0.789	0.855
								UH3	0.491	0.708	0.664	0.798	0.782	0.856